

Hypothesis 6: Does Fatigue Slow People Down Over Time?

Checking if Reaction Time and Accuracy Change Across the Experiment

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1 What is Hypothesis 6?

1.1 The Simple Idea

Think about any long task you have done — like sitting through a 3-hour exam. By the end, your brain is tired. You might take longer to answer questions or make more mistakes than at the start. This is called **cognitive fatigue**.

In our experiment, each participant sat through roughly **30 minutes** of reading and recognising sentences — hundreds of trials in total.

The question for H6 is simple:

Do people get slower or less accurate as the experiment goes on?

If yes, then fatigue is a real problem in this experiment. It means the results from H1 to H5 could be slightly biased — the differences we see might partly be caused by tiredness, not just the sentence type or voice.

1.2 How We Test It

We split each participant’s experiment into **3 equal time blocks**:

Block	When	What it represents
Block 1: Early	First third of trials	Fresh, not yet tired
Block 2: Middle	Middle third of trials	Settling in
Block 3: Late	Last third of trials	Potentially fatigued

We then check **two things**: 1. **Reaction Time (RT)** — Does it take longer to press the key later in the experiment? 2. **Accuracy** — Do people make more mistakes later in the experiment?

1.3 Why This Matters

If fatigue is present: - Future studies should add “trial order” as a variable in the model - Results from H1-H4 should be interpreted with caution - Experimenters should consider giving breaks

If fatigue is NOT present: - The experimental design is robust - Participants stayed focused throughout - We can be more confident in H1-H4 findings

2 Formal Hypothesis

2.1 In Plain Words

Null Hypothesis (H0): Reaction time and accuracy do NOT change significantly across the three time blocks (Early, Middle, Late). People perform the same throughout the experiment.

Alternative Hypothesis (H6): Reaction time increases and/or accuracy decreases significantly from Early to Late blocks, showing that cognitive fatigue builds up over the course of the experiment.

2.2 The Tests We Use

Because the same person appears in all three blocks, this is a **within-subjects (repeated measures)** design. We use:

Outcome	Test	Why
Reaction Time (RT)	Friedman Test	RT is skewed; this is the non-parametric repeated-measures test
Accuracy	Friedman Test	Same reason — accuracy can be skewed and bounded between 0 and 1

If the Friedman test is significant ($p < 0.05$), we run **post-hoc pairwise Wilcoxon tests** to find out which blocks differ from each other.

3 Step A: Reading and Splitting the Data

3.1 Loading the Logs

We read all participant log files, remove practice trials, and keep only the events where a participant pressed a key to say they recognised a sentence (IR pressed events).

But first — before we look at any timing data — we must check **which participants were actually paying attention**. This is called the validation step.

4 Step 0: Participant Validation (Who Passes the Quality Check?)

4.1 Why Do We Need This?

Not every participant should be included in the analysis. Some people may have:

- Clicked the spacebar randomly without really reading
- Zoned out and missed most of the sentences
- Not understood the task properly

To catch these people, the experiment included special **validation sentences**. These are easy, obviously repeated sentences that anyone paying attention should recognise. If a participant cannot pass these simple checks, their data is removed.

4.2 The Validation Formula

$$\text{Correct Hits} > \frac{\text{Wrong IR Presses}}{2} + \text{Missed IRs}$$

Where:

- **Correct Hits** = validation repeats where the person pressed spacebar (correct response)
- **Wrong IR Presses** = non-repeats where they pressed spacebar (false alarm)
- **Missed IRs** = validation repeats where they did NOT press spacebar (missed it)

In simple words: **your correct responses must clearly outnumber your mistakes.**

```
log_files <- list.files(log_dir, pattern = "\\log$", full.names = TRUE)
cat(sprintf("Found %d log files.\n", length(log_files)))
```

```
## Found 114 log files.
```

```
# Load ALL rows from all files (we need isValidation and isRepeat columns)
raw_all <- log_files %>%
  map_dfr(~ {
    tryCatch(
      read_csv(., show_col_types = FALSE, col_types = cols(.default = "c")),
      error = function(e) tibble()
    )
  }) %>%
  mutate(
    participant_ID = as.integer(participant_ID),
    Timestamp      = suppressWarnings(as.numeric(Timestamp)),
    `Accuracy IR`  = suppressWarnings(as.numeric(`Accuracy IR`)),
    Reaction_time_IR = suppressWarnings(as.numeric(Reaction_time_IR)),
    isRepeat       = coalesce(tolower(trimws(isRepeat)) == "true", FALSE),
    isValidation   = coalesce(tolower(trimws(isValidation)) == "true", FALSE)
  ) %>%
  arrange(participant_ID, Timestamp)

cat(sprintf("Total rows loaded: %s across %d participants.\n",
  format(nrow(raw_all), big.mark = ","),
  n_distinct(raw_all$participant_ID)))
```

```
## Total rows loaded: 81,329 across 114 participants.
```

```
# Remove practice rows and assign trial IDs (sorted by Timestamp = real order)
raw_trials <- raw_all %>%
  filter(!str_starts(Event, "Practice") | is.na(Event)) %>%
  arrange(participant_ID, Timestamp) %>%      # IMPORTANT: must be chronological
```

```

group_by(participant_ID) %>%
mutate(trial_id = cumsum(Event == "Sentence shown")) %>%
ungroup() %>%
filter(trial_id > 0)

# Get sentence metadata (one row per trial)
# isValidation can be flagged via the COLUMN or via the EVENT STRING
# This matches exactly what H3/H4 does: is_validation = isValidation | has_validation
sentence_meta <- raw_trials %>%
  filter(Event == "Sentence shown") %>%
  group_by(participant_ID, trial_id) %>%
  slice(1) %>%
  ungroup() %>%
  select(participant_ID, trial_id, isRepeat, isValidation)

# Check if "Validation" appears anywhere in the event string for each trial
has_validation_flag <- raw_trials %>%
  group_by(participant_ID, trial_id) %>%
  summarise(
    has_validation = any(str_detect(Event, "Validation"), na.rm = TRUE),
    ir_pressed      = any(str_detect(Event, "IR pressed"), na.rm = TRUE),
    .groups        = "drop"
  )

# Combine - a trial is a validation trial if EITHER the column OR event string says so
# This is identical to the H1-H4 pipeline
all_trials <- sentence_meta %>%
  inner_join(has_validation_flag, by = c("participant_ID", "trial_id")) %>%
  mutate(is_validation = isValidation | has_validation) # <-- key fix

# Keep only the validation trials
val_trials <- all_trials %>% filter(is_validation == TRUE)

cat(sprintf("Total validation trials found: %d\n", nrow(val_trials)))

## Total validation trials found: 5502

# Apply the validation formula:
# Correct Hits > (False Alarms / 2) + Missed
validation_summary <- val_trials %>%
  group_by(participant_ID) %>%
  summarise(
    Correct_Hits = sum( isRepeat == TRUE & ir_pressed == TRUE, na.rm = TRUE),
    False_Alarms = sum( isRepeat == FALSE & ir_pressed == TRUE, na.rm = TRUE),
    Missed       = sum( isRepeat == TRUE & ir_pressed == FALSE, na.rm = TRUE),
    .groups      = "drop"
  ) %>%

```

```

mutate(
  Passes = Correct_Hits > (False_Alarms / 2) + Missed,
  Status = ifelse(Passes, "Kept", "Excluded")
)

valid_ids    <- validation_summary %>% filter( Passes) %>% pull(participant_ID)
excluded_ids <- validation_summary %>% filter(!Passes) %>% pull(participant_ID)

cat(sprintf("Total participants:    %d\n", nrow(validation_summary)))

## Total participants:    114

cat(sprintf("Participants KEPT:      %d\n", length(valid_ids)))

## Participants KEPT:    112

cat(sprintf("Participants EXCLUDED: %d\n", length(excluded_ids)))

## Participants EXCLUDED: 2

if (length(excluded_ids) > 0) {
  cat(sprintf("Excluded IDs: %s\n", paste(excluded_ids, collapse = ", ")))
}

## Excluded IDs: 271, 299

```

4.3 Validation Results Table

Green rows = participant is KEPT. Red rows = participant is EXCLUDED.

```

validation_summary %>%
  select(participant_ID, Correct_Hits, False_Alarms, Missed, Status) %>%
  kable(
    caption = "Table 0. Participant Validation Results (who passed the quality check?)",
    col.names = c("Participant", "Correct Hits", "Wrong IR Presses",
                  "Missed IRs", "Status"),
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position", "scale_down"),
    bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE,
    position = "center"
  )

```

```

) %>%
column_spec(5,
  bold      = TRUE,
  color     = ifelse(validation_summary$Passes, "#1B5E20", "#B71C1C"),
  background = ifelse(validation_summary$Passes, "#E8F5E9", "#FDECEA")
) %>%
row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")

```

Table 3: Table 0. Participant Validation Results (who passed the quality check?)

Participant	Correct Hits	Wrong IR Presses	Missed IRs	Status
232	30	16	0	Kept
235	29	22	1	Kept
236	30	15	0	Kept
241	29	24	1	Kept
242	27	3	3	Kept
243	30	7	0	Kept
245	21	19	9	Kept
246	27	7	3	Kept
247	26	4	4	Kept
249	29	26	1	Kept
251	30	44	0	Kept
252	30	10	0	Kept
253	30	15	0	Kept
254	28	32	2	Kept
256	28	20	2	Kept
257	26	14	4	Kept
258	24	8	6	Kept
259	30	29	0	Kept
260	28	8	2	Kept
261	28	26	2	Kept
262	27	6	3	Kept
263	28	17	2	Kept
264	30	14	0	Kept
266	30	13	0	Kept
267	30	22	0	Kept
269	29	5	1	Kept
270	24	1	6	Kept
271	30	73	0	Excluded
273	27	6	3	Kept
276	29	13	1	Kept
277	30	25	0	Kept
279	30	10	0	Kept

280	27	11	3	Kept
281	29	46	1	Kept
282	30	26	0	Kept
283	25	6	5	Kept
285	30	6	0	Kept
286	29	13	1	Kept
289	30	15	0	Kept
290	29	10	1	Kept
292	29	43	1	Kept
293	30	9	0	Kept
294	29	37	1	Kept
295	30	5	0	Kept
296	28	13	2	Kept
297	30	42	0	Kept
298	26	10	4	Kept
299	13	20	17	Excluded
302	28	22	2	Kept
303	28	6	2	Kept
304	28	21	2	Kept
305	30	16	0	Kept
307	30	21	0	Kept
308	30	34	0	Kept
309	21	5	9	Kept
310	26	11	4	Kept
311	30	15	0	Kept
312	29	18	1	Kept
313	30	48	0	Kept
314	29	16	1	Kept
316	30	18	0	Kept
317	27	27	3	Kept
318	30	15	0	Kept
322	27	26	3	Kept
323	30	7	0	Kept
325	28	13	2	Kept
326	29	16	1	Kept
327	30	12	0	Kept
328	29	16	1	Kept
329	28	11	2	Kept
330	28	26	2	Kept
331	27	7	3	Kept
333	30	43	0	Kept
339	30	38	0	Kept
340	30	8	0	Kept
342	30	45	0	Kept

343	30	33	0	Kept
344	30	8	0	Kept
346	27	46	3	Kept
350	30	9	0	Kept
351	30	25	0	Kept
354	28	10	2	Kept
355	30	8	0	Kept
356	29	14	1	Kept
361	30	11	0	Kept
362	26	12	4	Kept
363	29	19	1	Kept
364	26	5	4	Kept
366	28	26	2	Kept
367	30	51	0	Kept
368	30	6	0	Kept
369	25	14	5	Kept
370	28	21	2	Kept
371	25	10	5	Kept
372	27	9	3	Kept
373	30	48	0	Kept
374	27	21	3	Kept
375	30	9	0	Kept
377	28	26	2	Kept
378	29	11	1	Kept
379	27	13	3	Kept
380	27	8	3	Kept
381	30	35	0	Kept
382	29	13	1	Kept
383	30	24	0	Kept
384	29	9	1	Kept
386	26	7	4	Kept
390	27	16	3	Kept
391	30	11	0	Kept
392	29	26	1	Kept
395	28	13	2	Kept
396	30	6	0	Kept
397	30	6	0	Kept
398	27	16	3	Kept

```
write_csv(validation_summary, file.path(output_dir, "H6_validation.csv"))
```

5 Step A: Loading Fatigue Data (Valid Participants Only)

5.1 Loading IR Press Events

Now that we know which participants passed the check, we load the IR (key-press) events for **valid participants only** and remove all practice rows.

```
# Read all log files, filter to valid participants + non-practice + IR presses
fatigue_raw <- log_files %>%
  map_dfr(~ {
    tryCatch({
      d <- read_csv(.x, show_col_types = FALSE, col_types = cols(.default = "c"))
      d %>%
        filter(
          !str_detect(Event, "Practice"),
          str_detect(Event, "IR pressed")
        )
    }, error = function(e) tibble())
  }) %>%
  mutate(
    participant_ID = as.integer(participant_ID),
    Reaction_time_IR = suppressWarnings(as.numeric(Reaction_time_IR)),
    `Accuracy IR` = suppressWarnings(as.numeric(`Accuracy IR`))
  ) %>%
  filter(
    !is.na(Reaction_time_IR),
    participant_ID %in% valid_ids    # Only keep validated participants
  )

cat(sprintf("IR press events (valid participants only): %s\n",
            format(nrow(fatigue_raw), big.mark = ",")))
```

```
## IR press events (valid participants only): 14,832
```

```
cat(sprintf("Valid participants in dataset: %d\n",
            n_distinct(fatigue_raw$participant_ID)))
```

```
## Valid participants in dataset: 112
```

5.2 Splitting into 3 Time Blocks

For each participant, we rank their trials from first to last (trial 1, 2, 3...), then split that sequence into three equal thirds.

```

# CRITICAL: Sort by Timestamp first so that row 1 = the FIRST event in real time
# Without this, block Early/Middle/Late would not reflect actual experiment order
fatigue_blocks <- fatigue_raw %>%
  arrange(participant_ID, Timestamp) %>%      # chronological order
  group_by(participant_ID) %>%
  mutate(
    trial_sequence = row_number(),             # 1 = first press, 2 = second, ...
    total_trials   = n(),
    block_id       = ntile(trial_sequence, 3)   # split into 3 equal thirds
  ) %>%
  mutate(
    Block_Name = case_when(
      block_id == 1 ~ "1: Early",
      block_id == 2 ~ "2: Middle",
      block_id == 3 ~ "3: Late"
    )
  ) %>%
  ungroup()

cat(sprintf(
  "Split %s chronologically-ordered events into 3 blocks across %d participants.\n",
  format(nrow(fatigue_blocks), big.mark = ","),
  n_distinct(fatigue_blocks$participant_ID)
))

```

```
## Split 14,832 chronologically-ordered events into 3 blocks across 112 participants.
```

```

# Sanity check: show min trial count per participant (should be at least ~3)
min_trials <- fatigue_blocks %>%
  group_by(participant_ID) %>%
  summarise(n = n(), .groups = "drop") %>%
  pull(n) %>% min()
cat(sprintf("Minimum IR trials per participant: %d\n", min_trials))

```

```
## Minimum IR trials per participant: 61
```

5.3 Preview Table 1 — Raw Rows with Block Assignment

This table shows you exactly what each row looks like AFTER the split, for **3 sample participants**. You can see the trial number (`trial_sequence`), the block it was assigned to (`block_id` and `Block_Name`), and the actual Reaction Time and Timestamp.

```

# Pick first 3 participants to show
sample_ids <- fatigue_blocks %>%
  distinct(participant_ID) %>%
  slice(1:3) %>%
  pull(participant_ID)

fatigue_blocks %>%
  filter(participant_ID %in% sample_ids) %>%
  mutate(
    # Timestamp is read as character - convert to numeric first, then to seconds
    Timestamp_s = round(suppressWarnings(as.numeric(Timestamp)) / 1000, 1)
  ) %>%
  select(
    Participant      = participant_ID,
    `Trial #`       = trial_sequence,
    `Total Trials`   = total_trials,
    `Block ID`       = block_id,
    `Block Name`     = Block_Name,
    `RT (ms)`        = Reaction_time_IR,
    `Timestamp (s)`  = Timestamp_s
  ) %>%
  kable(
    caption = "Preview Table 1. Sample rows showing how each trial was numbered and assigned to a block.",
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position", "scale_down"),
    bootstrap_options = c("striped", "hover", "condensed"),
    full_width      = FALSE,
    position        = "center",
    font_size       = 10
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")

```

Table 4: Preview Table 1. Sample rows showing how each trial was numbered and assigned to a block.

Participant	Trial #	Total Trials	Block ID	Block Name	RT (ms)	Timestamp (s)
232	1	139	1	1: Early	604	1770219420
232	2	139	1	1: Early	604	1770219420
232	3	139	1	1: Early	635	1770219446
232	4	139	1	1: Early	635	1770219446
232	5	139	1	1: Early	1300	1770219448
232	6	139	1	1: Early	1300	1770219448
232	7	139	1	1: Early	770	1770219480
232	8	139	1	1: Early	770	1770219480
232	9	139	1	1: Early	865	1770219497

232	10	139	1	1: Early	752	1770219515
232	11	139	1	1: Early	752	1770219515
232	12	139	1	1: Early	802	1770219518
232	13	139	1	1: Early	880	1770219520
232	14	139	1	1: Early	1077	1770219523
232	15	139	1	1: Early	1039	1770219527
232	16	139	1	1: Early	1039	1770219527
232	17	139	1	1: Early	880	1770219537
232	18	139	1	1: Early	880	1770219537
232	19	139	1	1: Early	2256	1770219544
232	20	139	1	1: Early	2256	1770219544
232	21	139	1	1: Early	1365	1770219547
232	22	139	1	1: Early	1365	1770219547
232	23	139	1	1: Early	868	1770219550
232	24	139	1	1: Early	827	1770219552
232	25	139	1	1: Early	656	1770219559
232	26	139	1	1: Early	656	1770219559
232	27	139	1	1: Early	564	1770219576
232	28	139	1	1: Early	564	1770219576
232	29	139	1	1: Early	945	1770219603
232	30	139	1	1: Early	1002	1770219607
232	31	139	1	1: Early	1002	1770219607
232	32	139	1	1: Early	826	1770219615
232	33	139	1	1: Early	810	1770219618
232	34	139	1	1: Early	855	1770219622
232	35	139	1	1: Early	855	1770219622
232	36	139	1	1: Early	665	1770219625
232	37	139	1	1: Early	712	1770219632
232	38	139	1	1: Early	712	1770219632
232	39	139	1	1: Early	1245	1770219651
232	40	139	1	1: Early	1245	1770219651
232	41	139	1	1: Early	832	1770219655
232	42	139	1	1: Early	2088	1770219659
232	43	139	1	1: Early	2088	1770219659
232	44	139	1	1: Early	799	1770219668
232	45	139	1	1: Early	799	1770219668
232	46	139	1	1: Early	1062	1770219676
232	47	139	1	1: Early	1096	1770219684
232	48	139	2	2: Middle	891	1770219749
232	49	139	2	2: Middle	1010	1770219753
232	50	139	2	2: Middle	767	1770219756
232	51	139	2	2: Middle	823	1770219797
232	52	139	2	2: Middle	823	1770219797
232	53	139	2	2: Middle	881	1770219808
232	54	139	2	2: Middle	688	1770219816
232	55	139	2	2: Middle	727	1770219845
232	56	139	2	2: Middle	727	1770219845
232	57	139	2	2: Middle	945	1770219848
232	58	139	2	2: Middle	945	1770219848

232	59	139	2	2: Middle	668	1770219860
232	60	139	2	2: Middle	668	1770219860
232	61	139	2	2: Middle	1657	1770219888
232	62	139	2	2: Middle	1657	1770219888
232	63	139	2	2: Middle	894	1770219891
232	64	139	2	2: Middle	806	1770219899
232	65	139	2	2: Middle	973	1770219913
232	66	139	2	2: Middle	973	1770219913
232	67	139	2	2: Middle	935	1770219930
232	68	139	2	2: Middle	1208	1770219939
232	69	139	2	2: Middle	1208	1770219939
232	70	139	2	2: Middle	756	1770219943
232	71	139	2	2: Middle	947	1770219953
232	72	139	2	2: Middle	947	1770219953
232	73	139	2	2: Middle	1097	1770219966
232	74	139	2	2: Middle	761	1770219970
232	75	139	2	2: Middle	761	1770219970
232	76	139	2	2: Middle	1598	1770219974
232	77	139	2	2: Middle	1598	1770219974
232	78	139	2	2: Middle	1151	1770219977
232	79	139	2	2: Middle	1151	1770219977
232	80	139	2	2: Middle	1156	1770219979
232	81	139	2	2: Middle	968	1770219982
232	82	139	2	2: Middle	886	1770219989
232	83	139	2	2: Middle	923	1770219991
232	84	139	2	2: Middle	923	1770219991
232	85	139	2	2: Middle	873	1770219993
232	86	139	2	2: Middle	873	1770219993
232	87	139	2	2: Middle	979	1770220004
232	88	139	2	2: Middle	952	1770220039
232	89	139	2	2: Middle	1996	1770220044
232	90	139	2	2: Middle	1996	1770220044
232	91	139	2	2: Middle	920	1770220073
232	92	139	2	2: Middle	1228	1770220076
232	93	139	2	2: Middle	1228	1770220076
232	94	139	3	3: Late	1611	1770220086
232	95	139	3	3: Late	1611	1770220086
232	96	139	3	3: Late	768	1770220089
232	97	139	3	3: Late	875	1770220092
232	98	139	3	3: Late	910	1770220173
232	99	139	3	3: Late	910	1770220173
232	100	139	3	3: Late	3837	1770220198
232	101	139	3	3: Late	3837	1770220198
232	102	139	3	3: Late	1078	1770220220
232	103	139	3	3: Late	1181	1770220223
232	104	139	3	3: Late	1181	1770220223
232	105	139	3	3: Late	935	1770220225
232	106	139	3	3: Late	737	1770220238
232	107	139	3	3: Late	737	1770220238

232	108	139	3	3: Late	757	1770220247
232	109	139	3	3: Late	1559	1770220250
232	110	139	3	3: Late	1559	1770220250
232	111	139	3	3: Late	1090	1770220253
232	112	139	3	3: Late	1090	1770220253
232	113	139	3	3: Late	1628	1770220271
232	114	139	3	3: Late	865	1770220284
232	115	139	3	3: Late	1092	1770220312
232	116	139	3	3: Late	1016	1770220330
232	117	139	3	3: Late	755	1770220334
232	118	139	3	3: Late	915	1770220337
232	119	139	3	3: Late	789	1770220344
232	120	139	3	3: Late	789	1770220344
232	121	139	3	3: Late	1233	1770220351
232	122	139	3	3: Late	3376	1770220365
232	123	139	3	3: Late	3376	1770220365
232	124	139	3	3: Late	751	1770220374
232	125	139	3	3: Late	751	1770220374
232	126	139	3	3: Late	867	1770220376
232	127	139	3	3: Late	867	1770220376
232	128	139	3	3: Late	1128	1770220399
232	129	139	3	3: Late	1128	1770220399
232	130	139	3	3: Late	955	1770220407
232	131	139	3	3: Late	955	1770220407
232	132	139	3	3: Late	962	1770220419
232	133	139	3	3: Late	797	1770220437
232	134	139	3	3: Late	797	1770220437
232	135	139	3	3: Late	878	1770220449
232	136	139	3	3: Late	1069	1770220462
232	137	139	3	3: Late	972	1770220468
232	138	139	3	3: Late	762	1770220470
232	139	139	3	3: Late	1022	1770220473
235	1	138	1	1: Early	1235	1770255731
235	2	138	1	1: Early	1235	1770255731
235	3	138	1	1: Early	925	1770255738
235	4	138	1	1: Early	925	1770255738
235	5	138	1	1: Early	1401	1770255747
235	6	138	1	1: Early	1401	1770255747
235	7	138	1	1: Early	1363	1770255750
235	8	138	1	1: Early	1002	1770255782
235	9	138	1	1: Early	1859	1770255800
235	10	138	1	1: Early	1859	1770255800
235	11	138	1	1: Early	828	1770255818
235	12	138	1	1: Early	828	1770255818
235	13	138	1	1: Early	973	1770255820
235	14	138	1	1: Early	1312	1770255828
235	15	138	1	1: Early	1312	1770255828
235	16	138	1	1: Early	2441	1770255832
235	17	138	1	1: Early	2441	1770255832

235	18	138	1	1: Early	2009	1770255836
235	19	138	1	1: Early	2009	1770255836
235	20	138	1	1: Early	1270	1770255846
235	21	138	1	1: Early	1019	1770255865
235	22	138	1	1: Early	1019	1770255865
235	23	138	1	1: Early	3162	1770255870
235	24	138	1	1: Early	3162	1770255870
235	25	138	1	1: Early	2083	1770255888
235	26	138	1	1: Early	2083	1770255888
235	27	138	1	1: Early	2861	1770255896
235	28	138	1	1: Early	2861	1770255896
235	29	138	1	1: Early	1078	1770255898
235	30	138	1	1: Early	1253	1770255916
235	31	138	1	1: Early	1253	1770255916
235	32	138	1	1: Early	1068	1770255919
235	33	138	1	1: Early	1068	1770255919
235	34	138	1	1: Early	2185	1770255922
235	35	138	1	1: Early	2185	1770255922
235	36	138	1	1: Early	1752	1770255927
235	37	138	1	1: Early	1752	1770255927
235	38	138	1	1: Early	2918	1770255932
235	39	138	1	1: Early	2144	1770255937
235	40	138	1	1: Early	2394	1770255941
235	41	138	1	1: Early	1070	1770255958
235	42	138	1	1: Early	1070	1770255958
235	43	138	1	1: Early	1254	1770255961
235	44	138	1	1: Early	1254	1770255961
235	45	138	1	1: Early	706	1770255969
235	46	138	1	1: Early	1070	1770255971
235	47	138	2	2: Middle	1070	1770255971
235	48	138	2	2: Middle	1674	1770255990
235	49	138	2	2: Middle	1674	1770255990
235	50	138	2	2: Middle	2925	1770255997
235	51	138	2	2: Middle	2925	1770255997
235	52	138	2	2: Middle	1591	1770256004
235	53	138	2	2: Middle	2402	1770256008
235	54	138	2	2: Middle	2402	1770256008
235	55	138	2	2: Middle	944	1770256010
235	56	138	2	2: Middle	944	1770256010
235	57	138	2	2: Middle	1379	1770256024
235	58	138	2	2: Middle	1166	1770256092
235	59	138	2	2: Middle	1166	1770256092
235	60	138	2	2: Middle	1339	1770256105
235	61	138	2	2: Middle	1339	1770256105
235	62	138	2	2: Middle	1243	1770256114
235	63	138	2	2: Middle	1081	1770256138
235	64	138	2	2: Middle	1081	1770256138
235	65	138	2	2: Middle	1759	1770256146
235	66	138	2	2: Middle	1043	1770256184

235	67	138	2	2: Middle	1043	1770256184
235	68	138	2	2: Middle	1895	1770256188
235	69	138	2	2: Middle	1895	1770256188
235	70	138	2	2: Middle	1297	1770256190
235	71	138	2	2: Middle	1203	1770256193
235	72	138	2	2: Middle	1505	1770256211
235	73	138	2	2: Middle	1505	1770256211
235	74	138	2	2: Middle	2716	1770256217
235	75	138	2	2: Middle	1730	1770256240
235	76	138	2	2: Middle	1730	1770256240
235	77	138	2	2: Middle	1452	1770256244
235	78	138	2	2: Middle	1452	1770256244
235	79	138	2	2: Middle	1507	1770256247
235	80	138	2	2: Middle	1507	1770256247
235	81	138	2	2: Middle	1361	1770256297
235	82	138	2	2: Middle	741	1770256330
235	83	138	2	2: Middle	858	1770256332
235	84	138	2	2: Middle	858	1770256332
235	85	138	2	2: Middle	786	1770256339
235	86	138	2	2: Middle	786	1770256339
235	87	138	2	2: Middle	1053	1770256356
235	88	138	2	2: Middle	1053	1770256356
235	89	138	2	2: Middle	1299	1770256364
235	90	138	2	2: Middle	1299	1770256364
235	91	138	2	2: Middle	1514	1770256368
235	92	138	2	2: Middle	878	1770256369
235	93	138	3	3: Late	878	1770256369
235	94	138	3	3: Late	920	1770256372
235	95	138	3	3: Late	735	1770256395
235	96	138	3	3: Late	735	1770256395
235	97	138	3	3: Late	929	1770256398
235	98	138	3	3: Late	1485	1770256400
235	99	138	3	3: Late	1937	1770256530
235	100	138	3	3: Late	1937	1770256530
235	101	138	3	3: Late	819	1770256567
235	102	138	3	3: Late	1640	1770256581
235	103	138	3	3: Late	1005	1770256583
235	104	138	3	3: Late	1005	1770256583
235	105	138	3	3: Late	2414	1770256587
235	106	138	3	3: Late	2169	1770256601
235	107	138	3	3: Late	1078	1770256625
235	108	138	3	3: Late	1078	1770256625
235	109	138	3	3: Late	964	1770256632
235	110	138	3	3: Late	964	1770256632
235	111	138	3	3: Late	792	1770256644
235	112	138	3	3: Late	792	1770256644
235	113	138	3	3: Late	977	1770256646
235	114	138	3	3: Late	1003	1770256655
235	115	138	3	3: Late	924	1770256657
235	116	138	3	3: Late	924	1770256657

235	117	138	3	3: Late	1453	1770256665
235	118	138	3	3: Late	1453	1770256665
235	119	138	3	3: Late	1606	1770256667
235	120	138	3	3: Late	2739	1770256672
235	121	138	3	3: Late	1414	1770256704
235	122	138	3	3: Late	2982	1770256714
235	123	138	3	3: Late	1325	1770256743
235	124	138	3	3: Late	1325	1770256743
235	125	138	3	3: Late	928	1770256746
235	126	138	3	3: Late	938	1770256769
235	127	138	3	3: Late	938	1770256769
235	128	138	3	3: Late	1372	1770256771
235	129	138	3	3: Late	1441	1770256780
235	130	138	3	3: Late	1441	1770256780
235	131	138	3	3: Late	850	1770256807
235	132	138	3	3: Late	850	1770256807
235	133	138	3	3: Late	901	1770256809
235	134	138	3	3: Late	901	1770256809
235	135	138	3	3: Late	819	1770256821
235	136	138	3	3: Late	819	1770256821
235	137	138	3	3: Late	2039	1770256831
235	138	138	3	3: Late	1305	1770256833
236	1	136	1	1: Early	1089	1770293083
236	2	136	1	1: Early	3148	1770293108
236	3	136	1	1: Early	2919	1770293116
236	4	136	1	1: Early	1465	1770293139
236	5	136	1	1: Early	1465	1770293139
236	6	136	1	1: Early	2128	1770293143
236	7	136	1	1: Early	2638	1770293147
236	8	136	1	1: Early	2306	1770293156
236	9	136	1	1: Early	2306	1770293156
236	10	136	1	1: Early	3126	1770293162
236	11	136	1	1: Early	1730	1770293165
236	12	136	1	1: Early	1730	1770293165
236	13	136	1	1: Early	1295	1770293173
236	14	136	1	1: Early	1147	1770293196
236	15	136	1	1: Early	1147	1770293196
236	16	136	1	1: Early	2798	1770293200
236	17	136	1	1: Early	2624	1770293204
236	18	136	1	1: Early	2624	1770293204
236	19	136	1	1: Early	2391	1770293208
236	20	136	1	1: Early	2391	1770293208
236	21	136	1	1: Early	1929	1770293221
236	22	136	1	1: Early	2799	1770293226
236	23	136	1	1: Early	2799	1770293226
236	24	136	1	1: Early	2651	1770293230
236	25	136	1	1: Early	1421	1770293243
236	26	136	1	1: Early	1421	1770293243
236	27	136	1	1: Early	1231	1770293246
236	28	136	1	1: Early	1231	1770293246

236	29	136	1	1: Early	1748	1770293284
236	30	136	1	1: Early	2217	1770293299
236	31	136	1	1: Early	1978	1770293304
236	32	136	1	1: Early	1978	1770293304
236	33	136	1	1: Early	2734	1770293308
236	34	136	1	1: Early	2734	1770293308
236	35	136	1	1: Early	1439	1770293311
236	36	136	1	1: Early	1439	1770293311
236	37	136	1	1: Early	1594	1770293314
236	38	136	1	1: Early	2694	1770293333
236	39	136	1	1: Early	2694	1770293333
236	40	136	1	1: Early	1245	1770293336
236	41	136	1	1: Early	1245	1770293336
236	42	136	1	1: Early	2188	1770293344
236	43	136	1	1: Early	2188	1770293344
236	44	136	1	1: Early	903	1770293346
236	45	136	1	1: Early	903	1770293346
236	46	136	1	1: Early	978	1770293359
236	47	136	2	2: Middle	928	1770293361
236	48	136	2	2: Middle	928	1770293361
236	49	136	2	2: Middle	2569	1770293364
236	50	136	2	2: Middle	1317	1770293522
236	51	136	2	2: Middle	1317	1770293522
236	52	136	2	2: Middle	2487	1770293565
236	53	136	2	2: Middle	1660	1770293570
236	54	136	2	2: Middle	1768	1770293573
236	55	136	2	2: Middle	1768	1770293573
236	56	136	2	2: Middle	873	1770293580
236	57	136	2	2: Middle	873	1770293580
236	58	136	2	2: Middle	2062	1770293593
236	59	136	2	2: Middle	2062	1770293593
236	60	136	2	2: Middle	2622	1770293612
236	61	136	2	2: Middle	2622	1770293612
236	62	136	2	2: Middle	786	1770293615
236	63	136	2	2: Middle	786	1770293615
236	64	136	2	2: Middle	1003	1770293617
236	65	136	2	2: Middle	1003	1770293617
236	66	136	2	2: Middle	1820	1770293621
236	67	136	2	2: Middle	2190	1770293649
236	68	136	2	2: Middle	739	1770293657
236	69	136	2	2: Middle	739	1770293657
236	70	136	2	2: Middle	2779	1770293666
236	71	136	2	2: Middle	931	1770293678
236	72	136	2	2: Middle	931	1770293678
236	73	136	2	2: Middle	1792	1770293681
236	74	136	2	2: Middle	3143	1770293711
236	75	136	2	2: Middle	3143	1770293711
236	76	136	2	2: Middle	1095	1770293713
236	77	136	2	2: Middle	820	1770293725
236	78	136	2	2: Middle	820	1770293725

236	79	136	2	2: Middle	1839	1770293728
236	80	136	2	2: Middle	1308	1770293731
236	81	136	2	2: Middle	1308	1770293731
236	82	136	2	2: Middle	1463	1770293734
236	83	136	2	2: Middle	2104	1770293742
236	84	136	2	2: Middle	2104	1770293742
236	85	136	2	2: Middle	1902	1770293751
236	86	136	2	2: Middle	943	1770293759
236	87	136	2	2: Middle	943	1770293759
236	88	136	2	2: Middle	1123	1770293771
236	89	136	2	2: Middle	932	1770293783
236	90	136	2	2: Middle	1608	1770293791
236	91	136	2	2: Middle	81	1770293797
236	92	136	3	3: Late	81	1770293797
236	93	136	3	3: Late	1803	1770293809
236	94	136	3	3: Late	1275	1770293813
236	95	136	3	3: Late	1275	1770293813
236	96	136	3	3: Late	1793	1770293821
236	97	136	3	3: Late	908	1770293956
236	98	136	3	3: Late	908	1770293956
236	99	136	3	3: Late	786	1770293963
236	100	136	3	3: Late	786	1770293963
236	101	136	3	3: Late	808	1770293989
236	102	136	3	3: Late	808	1770293989
236	103	136	3	3: Late	1090	1770294001
236	104	136	3	3: Late	1835	1770294014
236	105	136	3	3: Late	944	1770294027
236	106	136	3	3: Late	1479	1770294030
236	107	136	3	3: Late	1933	1770294038
236	108	136	3	3: Late	1059	1770294050
236	109	136	3	3: Late	1059	1770294050
236	110	136	3	3: Late	973	1770294053
236	111	136	3	3: Late	2220	1770294077
236	112	136	3	3: Late	2220	1770294077
236	113	136	3	3: Late	962	1770294083
236	114	136	3	3: Late	962	1770294083
236	115	136	3	3: Late	1793	1770294086
236	116	136	3	3: Late	1793	1770294086
236	117	136	3	3: Late	1412	1770294089
236	118	136	3	3: Late	1763	1770294132
236	119	136	3	3: Late	3006	1770294139
236	120	136	3	3: Late	3006	1770294139
236	121	136	3	3: Late	2666	1770294145
236	122	136	3	3: Late	2493	1770294154
236	123	136	3	3: Late	1335	1770294158
236	124	136	3	3: Late	1335	1770294158
236	125	136	3	3: Late	1590	1770294161
236	126	136	3	3: Late	1014	1770294169
236	127	136	3	3: Late	2707	1770294198
236	128	136	3	3: Late	2707	1770294198

236	129	136	3	3: Late	965	1770294205
236	130	136	3	3: Late	965	1770294205
236	131	136	3	3: Late	1077	1770294227
236	132	136	3	3: Late	1161	1770294234
236	133	136	3	3: Late	1161	1770294234
236	134	136	3	3: Late	1238	1770294256
236	135	136	3	3: Late	1543	1770294259
236	136	136	3	3: Late	976	1770294261

5.4 Preview Table 2 — Block Boundaries Per Participant

This table shows, for **each participant in each block**: - The **first** trial number in that block (where Early starts) - The **last** trial number in that block (where Late ends) - How many trials are in the block

This proves the splitting was done correctly on a per-person basis.

```
block_boundaries <- fatigue_blocks %>%
  mutate(Timestamp_num = suppressWarnings(as.numeric(Timestamp))) %>%
  group_by(participant_ID, Block_Name) %>%
  summarise(
    First_Trial      = min(trial_sequence),
    Last_Trial       = max(trial_sequence),
    N_Trials         = n(),
    First_Timestamp_s = round(min(Timestamp_num, na.rm = TRUE) / 1000, 0),
    Last_Timestamp_s  = round(max(Timestamp_num, na.rm = TRUE) / 1000, 0),
    .groups          = "drop"
  ) %>%
  arrange(participant_ID, Block_Name)

# Show first 3 participants only (full table would be too large)
block_boundaries %>%
  filter(participant_ID %in% sample_ids) %>%
  kable(
    caption = "Preview Table 2. Block boundaries for 3 sample participants. Shows exactly wh",
    col.names = c("Participant", "Block", "First Trial#",
                  "Last Trial#", "N Trials",
                  "Block Start (s)", "Block End (s)"),
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE,
    position = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white") %>%
  pack_rows("Participant 1", 1, 3) %>%
```

```
pack_rows("Participant 2", 4, 6) %>%
pack_rows("Participant 3", 7, 9)
```

Table 5: Preview Table 2. Block boundaries for 3 sample participants. Shows exactly which trial numbers and times-tamps fall into Early, Middle, and Late.

Participant	Block	First Trial#	Last Trial#	N Trials	Block Start (s)	Block End (s)
Participant 1						
232	1: Early	1	47	47	1770219420	1770219684
232	2: Middle	48	93	46	1770219749	1770220076
232	3: Late	94	139	46	1770220086	1770220473
Participant 2						
235	1: Early	1	46	46	1770255731	1770255971
235	2: Middle	47	92	46	1770255971	1770256369
235	3: Late	93	138	46	1770256369	1770256833
Participant 3						
236	1: Early	1	46	46	1770293083	1770293359
236	2: Middle	47	91	45	1770293361	1770293797
236	3: Late	92	136	45	1770293797	1770294261

5.5 Preview Table 3 — Full Block Size Summary (All Participants)

This table shows every participant's trial count in each block. Blocks should be roughly equal in size (within 1 trial of each other).

```
block_sizes <- fatigue_blocks %>%
  group_by(participant_ID, Block_Name) %>%
  summarise(N_Trials = n(), .groups = "drop") %>%
  pivot_wider(
    names_from = Block_Name,
    values_from = N_Trials
  ) %>%
  mutate(
    Total = `1: Early` + `2: Middle` + `3: Late`,
    Max_N = pmax(`1: Early`, `2: Middle`, `3: Late`),
    Min_N = pmin(`1: Early`, `2: Middle`, `3: Late`),
    Balanced = ifelse(Max_N - Min_N <= 1, "Yes", "No")
  ) %>%
  select(-Max_N, -Min_N)

block_sizes %>%
  kable(
    caption = "Preview Table 3. Number of trials per block per participant. 'Balanced = Yes'"
    col.names = c("Participant", "Early", "Middle", "Late",
```

```

        "Total Trials", "Balanced?"),
    booktabs = TRUE
) %>%
kable_styling(
  latex_options = c("striped", "HOLD_position", "scale_down"),
  bootstrap_options = c("striped", "hover", "condensed"),
  full_width = FALSE,
  position = "center",
  font_size = 10
) %>%
row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white") %>%
column_spec(6,
  bold = TRUE,
  color = ifelse(block_sizes$Balanced == "Yes", "#1B5E20", "#B71C1C"),
  background = ifelse(block_sizes$Balanced == "Yes", "#E8F5E9", "#FDECEA")
)

```

Table 6: Preview Table 3. Number of trials per block per participant. 'Balanced = Yes' means the three blocks are equal (or differ by at most 1 trial).

Participant	Early	Middle	Late	Total Trials	Balanced?
232	47	46	46	139	Yes
235	46	46	46	138	Yes
236	46	45	45	136	Yes
241	50	50	49	149	Yes
242	34	33	33	100	Yes
243	40	40	40	120	Yes
245	33	33	32	98	Yes
246	31	30	30	91	Yes
247	34	34	34	102	Yes
249	49	48	48	145	Yes
251	65	65	65	195	Yes
252	42	42	41	125	Yes
253	46	46	46	138	Yes
254	53	52	52	157	Yes
256	47	47	47	141	Yes
257	36	36	35	107	Yes
258	35	35	35	105	Yes
259	52	52	52	156	Yes
260	37	37	37	111	Yes
261	52	52	52	156	Yes
262	35	35	35	105	Yes
263	42	41	41	124	Yes
264	45	44	44	133	Yes
266	43	43	42	128	Yes
267	50	50	50	150	Yes
269	37	36	36	109	Yes

270	28	28	28	84	Yes
273	32	32	31	95	Yes
276	40	40	39	119	Yes
277	53	52	52	157	Yes
279	42	42	41	125	Yes
280	41	41	41	123	Yes
281	64	63	63	190	Yes
282	52	52	52	156	Yes
283	30	29	29	88	Yes
285	40	39	39	118	Yes
286	38	37	37	112	Yes
289	44	44	43	131	Yes
290	41	41	41	123	Yes
292	63	63	62	188	Yes
293	40	40	40	120	Yes
294	59	59	59	177	Yes
295	38	38	38	114	Yes
296	42	42	41	125	Yes
297	63	63	62	188	Yes
298	34	34	34	102	Yes
302	46	46	45	137	Yes
303	37	37	36	110	Yes
304	45	44	44	133	Yes
305	47	47	46	140	Yes
307	50	50	50	150	Yes
308	57	57	57	171	Yes
309	21	20	20	61	Yes
310	38	38	38	114	Yes
311	45	45	44	134	Yes
312	46	46	45	137	Yes
313	68	67	67	202	Yes
314	44	44	44	132	Yes
316	47	47	46	140	Yes
317	50	49	49	148	Yes
318	45	45	44	134	Yes
322	49	48	48	145	Yes
323	40	40	40	120	Yes
325	38	37	37	112	Yes
326	43	42	42	127	Yes
327	43	43	43	129	Yes
328	44	44	44	132	Yes
329	39	39	39	117	Yes
330	50	49	49	148	Yes
331	35	35	35	105	Yes
333	64	64	64	192	Yes
339	61	61	60	182	Yes
340	41	41	41	123	Yes
342	66	66	65	197	Yes
343	53	53	53	159	Yes

344	41	41	40	122	Yes
346	64	63	63	190	Yes
350	39	39	38	116	Yes
351	52	52	51	155	Yes
354	38	38	38	114	Yes
355	40	40	40	120	Yes
356	44	43	43	130	Yes
361	43	43	43	129	Yes
362	37	37	36	110	Yes
363	46	46	45	137	Yes
364	33	33	32	98	Yes
366	52	51	51	154	Yes
367	70	70	69	209	Yes
368	37	37	36	110	Yes
369	38	37	37	112	Yes
370	45	45	45	135	Yes
371	30	30	30	90	Yes
372	36	36	35	107	Yes
373	64	64	63	191	Yes
374	42	42	42	126	Yes
375	40	40	39	119	Yes
377	49	49	48	146	Yes
378	43	42	42	127	Yes
379	35	35	35	105	Yes
380	38	38	37	113	Yes
381	60	59	59	178	Yes
382	43	43	43	129	Yes
383	51	50	50	151	Yes
384	39	39	39	117	Yes
386	32	32	32	96	Yes
390	39	39	38	116	Yes
391	41	40	40	121	Yes
392	51	51	51	153	Yes
395	41	40	40	121	Yes
396	39	39	38	116	Yes
397	39	39	38	116	Yes
398	43	43	43	129	Yes

```
cat(sprintf("\nAll participants balanced: %s\n",
           ifelse(all(block_sizes$Balanced == "Yes"), "YES", "NO -- check data!")))
```

```
##
## All participants balanced: YES
```

5.6 Computing One Score Per Person Per Block

We calculate each participant's **median RT** and **mean accuracy** within each block. Using the **median** for RT is important because reaction times are often skewed by occasional very slow responses (due to momentary distractions), and the median is less affected by these outliers.

```
person_blocks <- fatigue_blocks %>%
  group_by(participant_ID, Block_Name) %>%
  summarise(
    Median_RT = median(Reaction_time_IR, na.rm = TRUE),
    Mean_Acc = mean(`Accuracy IR`, na.rm = TRUE),
    n_trials = n(),
    .groups = "drop"
  ) %>%
  mutate(Block_Name = factor(Block_Name,
                             levels = c("1: Early", "2: Middle", "3: Late")))

write_csv(person_blocks, file.path(output_dir, "H6_block_scores.csv"))

cat(sprintf("Block scores ready: %d rows (%d participants x 3 blocks)\n",
           nrow(person_blocks), n_distinct(person_blocks$participant_ID)))
```

```
## Block scores ready: 336 rows (112 participants x 3 blocks)
```

6 Step B: Descriptive Statistics

Before running any statistical tests, let us look at the numbers first.

6.1 Reaction Time by Block

```
rt_desc <- person_blocks %>%
  group_by(Block_Name) %>%
  summarise(
    N = n(),
    Mean_RT = round(mean(Median_RT, na.rm = TRUE), 1),
    Median_RT = round(median(Median_RT, na.rm = TRUE), 1),
    SD_RT = round(sd(Median_RT, na.rm = TRUE), 1),
    Min_RT = round(min(Median_RT, na.rm = TRUE), 1),
    Max_RT = round(max(Median_RT, na.rm = TRUE), 1),
    .groups = "drop"
  )
```

```

rt_desc %>%
  kable(
    caption = "Table 1. Reaction Time (ms) by Time Block",
    col.names = c("Block", "N", "Mean (ms)", "Median (ms)", "SD", "Min", "Max"),
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width = FALSE,
    position = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")

```

Table 7: Table 1. Reaction Time (ms) by Time Block

Block	N	Mean (ms)	Median (ms)	SD	Min	Max
1: Early	112	1539.4	1466.2	NA	1466.2	1466.2
2: Middle	112	1502.0	1480.5	NA	1480.5	1480.5
3: Late	112	1451.2	1414.0	NA	1414.0	1414.0

6.2 Accuracy by Block

```

acc_desc <- person_blocks %>%
  group_by(Block_Name) %>%
  summarise(
    N = n(),
    Mean_Acc = round(mean(Mean_Acc, na.rm = TRUE), 4),
    SD_Acc = round(sd(Mean_Acc, na.rm = TRUE), 4),
    Min_Acc = round(min(Mean_Acc, na.rm = TRUE), 4),
    Max_Acc = round(max(Mean_Acc, na.rm = TRUE), 4),
    .groups = "drop"
  )

acc_desc %>%
  kable(
    caption = "Table 2. Accuracy (proportion correct) by Time Block",
    col.names = c("Block", "N", "Mean Accuracy", "SD", "Min", "Max"),
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width = FALSE,

```

```

    position          = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")

```

Table 8: Table 2. Accuracy (proportion correct) by Time Block

Block	N	Mean Accuracy	SD	Min	Max
1: Early	112	0.6743	NA	0.6743	0.6743
2: Middle	112	0.7571	NA	0.7571	0.7571
3: Late	112	0.8267	NA	0.8267	0.8267

7 Step C: Visualising the Data

Before running the tests, we look at the data visually. This helps us spot patterns that numbers alone might hide.

7.1 Figure 1 - Reaction Time Across Blocks

```

p_rt_violin <- ggplot(person_blocks,
  aes(x = Block_Name, y = Median_RT, fill = Block_Name)) +
  geom_violin(alpha = 0.55, colour = "grey60", width = 0.8) +
  geom_boxplot(width = 0.18, alpha = 0.85, outlier.shape = NA, colour = "grey20") +
  geom_jitter(width = 0.07, alpha = 0.35, size = 1.5, colour = "grey30") +
  scale_fill_manual(values = PAL_BLOCKS, guide = "none") +
  scale_y_continuous(limits = c(0, 3200), oob = scales::squish,
    labels = comma_format()) +
  labs(
    title = "Figure 1: Reaction Time by Time Block",
    subtitle = "Y-axis starts at 0 for honest comparison. If people get slower, boxes shift up",
    x = "Time Block",
    y = "Median Reaction Time per Participant (ms)"
  )
p_rt_violin

```

```

ggsave(file.path(output_dir, "H6_RT_violin.png"),
  p_rt_violin, width = 7, height = 5, dpi = 150)

```

Figure 1: Reaction Time by Time Block

Y-axis starts at 0 for honest comparison. If people get slower, boxes shift up.

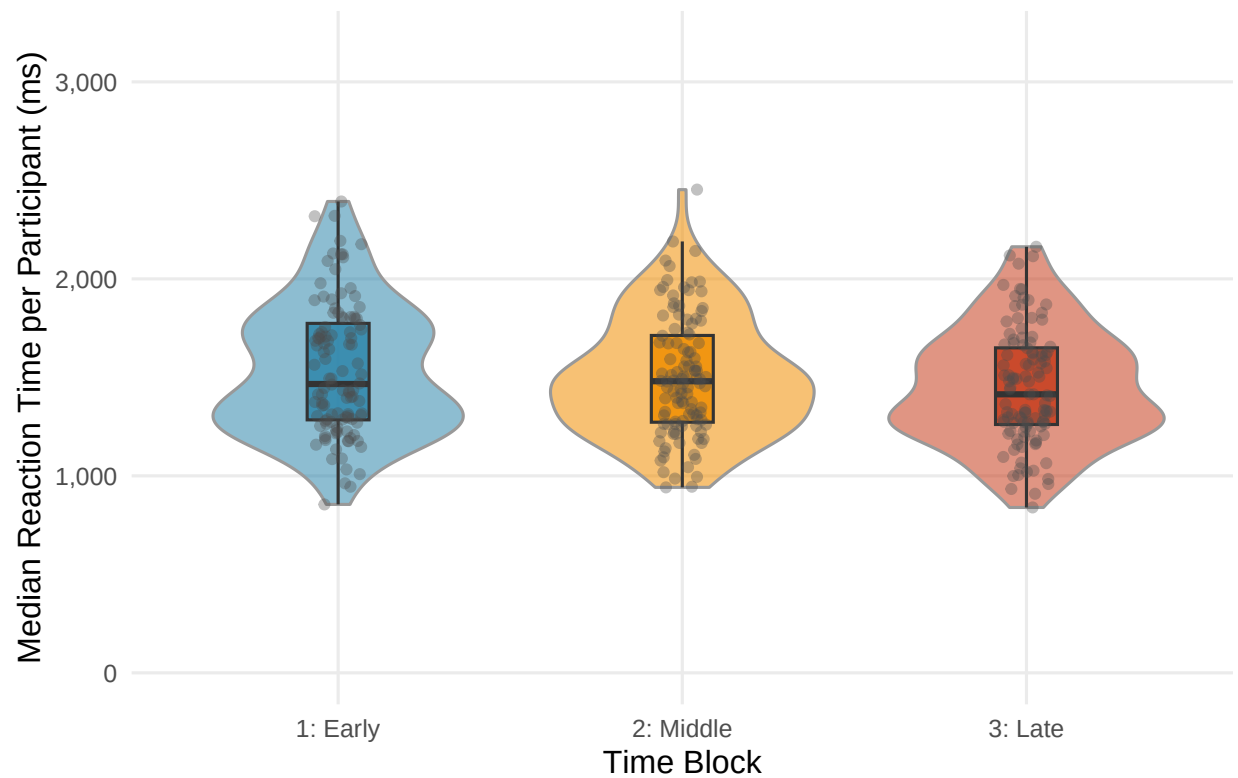


Figure 1: Figure 1. Reaction time distributions across the three time blocks. Each dot = one participant. The box shows the median and spread. If people get slower, the boxes should shift upward from Early to Late.

7.2 Figure 2 - Accuracy Across Blocks

```
p_acc_violin <- ggplot(person_blocks,
  aes(x = Block_Name, y = Mean_Acc, fill = Block_Name)) +
  geom_violin(alpha = 0.55, colour = "grey60", width = 0.8) +
  geom_boxplot(width = 0.18, alpha = 0.85, outlier.shape = NA, colour = "grey20") +
  geom_jitter(width = 0.07, alpha = 0.35, size = 1.5, colour = "grey30") +
  scale_fill_manual(values = PAL_BLOCKS, guide = "none") +
  scale_y_continuous(limits = c(0, 1), labels = percent_format()) +
  labs(
    title = "Figure 2: Accuracy by Time Block",
    subtitle = "Does accuracy drop from Early to Late in the experiment?",
    x = "Time Block",
    y = "Mean Accuracy per Participant (proportion)"
  )
p_acc_violin
```

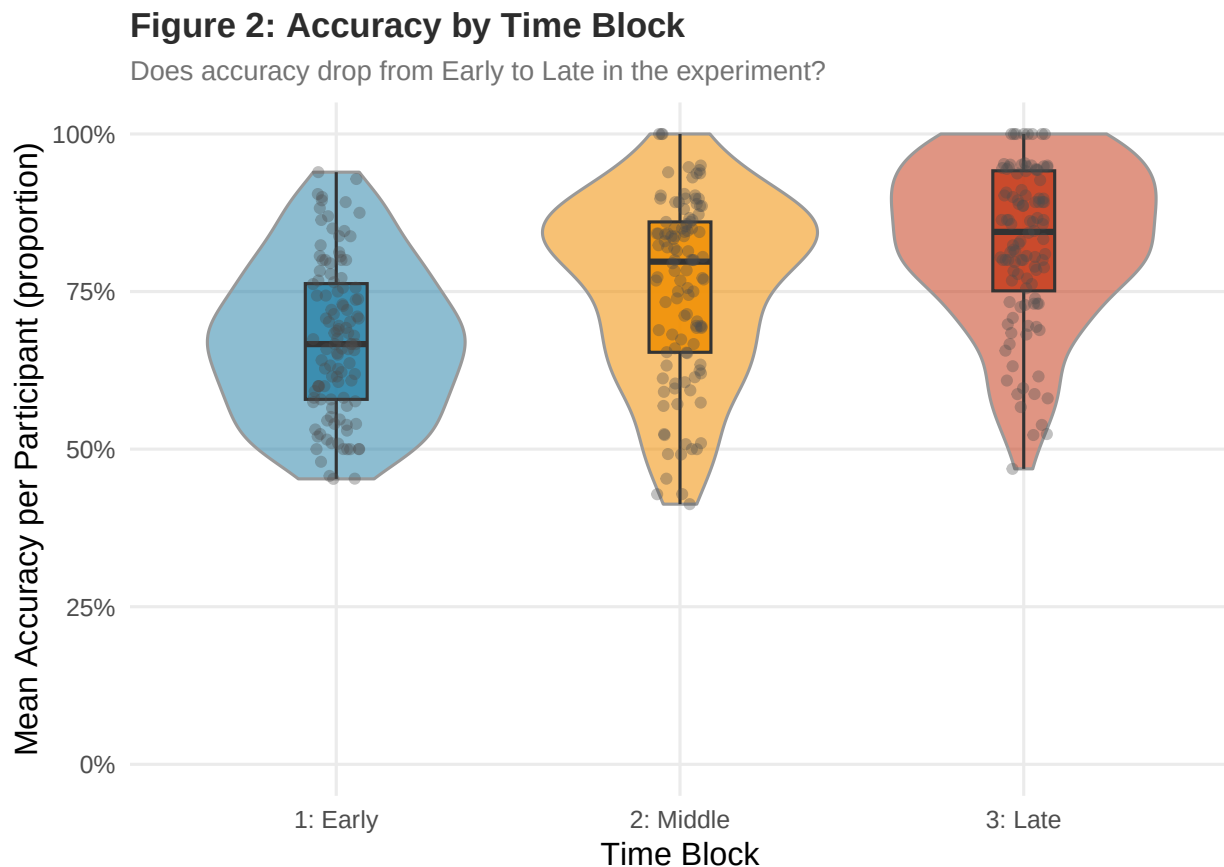


Figure 2: Figure 2. Accuracy distributions across the three time blocks. Each dot = one participant. If people make more mistakes later, the boxes should shift downward from Early to Late.

```
ggsave(file.path(output_dir, "H6_Acc_violin.png"),
       p_acc_violin, width = 7, height = 5, dpi = 150)
```

7.3 Figure 3 - Trend Line Across Blocks

This plot shows the **average across all participants** at each block, with error bars showing how much participants varied. A clear upward trend for RT (or downward for accuracy) would suggest fatigue.

```
# Summarise for trend plot
trend_rt <- person_blocks %>%
  group_by(Block_Name) %>%
  summarise(
    Mean = mean(Median_RT, na.rm = TRUE),
    SE    = sd(Median_RT, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

trend_acc <- person_blocks %>%
  group_by(Block_Name) %>%
  summarise(
    Mean = mean(Mean_Acc, na.rm = TRUE),
    SE    = sd(Mean_Acc, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

p_trend_rt <- ggplot(trend_rt, aes(x = Block_Name, y = Mean, group = 1)) +
  geom_line(colour = PAL_BLOCKS["1: Early"], linewidth = 1.2) +
  geom_point(aes(colour = Block_Name), size = 4) +
  geom_errorbar(aes(ymin = Mean - SE, ymax = Mean + SE),
               width = 0.15, colour = "grey40", linewidth = 0.8) +
  scale_colour_manual(values = PAL_BLOCKS, guide = "none") +
  scale_y_continuous(limits = c(0, NA), labels = comma_format()) +
  labs(
    title    = "Fig 3a: Average RT Trend Across Blocks",
    subtitle = "Y-axis starts at 0. Each point = group mean. Bars = +/- 1 standard error.",
    x        = "Time Block",
    y        = "Mean Median RT (ms)"
  )

p_trend_acc <- ggplot(trend_acc, aes(x = Block_Name, y = Mean, group = 1)) +
  geom_line(colour = PAL_BLOCKS["1: Early"], linewidth = 1.2) +
  geom_point(aes(colour = Block_Name), size = 4) +
  geom_errorbar(aes(ymin = Mean - SE, ymax = Mean + SE),
               width = 0.15, colour = "grey40", linewidth = 0.8) +
  scale_colour_manual(values = PAL_BLOCKS, guide = "none") +
```



```

scale_y_continuous(limits = c(0, 1), labels = percent_format()) +
labs(
  title      = "Fig 3b: Average Accuracy Trend Across Blocks",
  subtitle   = "Each point = group mean. Bars = standard error.",
  x          = "Time Block",
  y          = "Mean Accuracy"
)

# Stack vertically - no clipping
p_trend_rt / p_trend_acc

```

```

ggsave(file.path(output_dir, "H6_trend_lines.png"),
        p_trend_rt / p_trend_acc, width = 7, height = 7, dpi = 150)

```

7.4 Figure 4 - Individual Participant Trajectories

This plot shows each participant as a separate line. It answers: do **most** people slow down, or just a few outliers?

```

p_spaghetti_rt <- ggplot(person_blocks,
  aes(x = Block_Name, y = Median_RT, group = participant_ID)) +
  geom_line(alpha = 0.25, colour = "#2E86AB", linewidth = 0.5) +
  stat_summary(aes(group = 1), fun = mean, geom = "line",
    colour = "#C73E1D", linewidth = 1.5) +
  stat_summary(aes(group = 1), fun = mean, geom = "point",
    colour = "#C73E1D", size = 3) +
  scale_y_continuous(limits = c(0, NA), labels = comma_format()) +
  labs(
    title      = "Fig 4a: Individual RT Trajectories",
    subtitle   = "Y-axis starts at 0. Thin lines = each participant. Thick red = group average.",
    x          = "Time Block",
    y          = "Median RT (ms)"
  )

p_spaghetti_acc <- ggplot(person_blocks,
  aes(x = Block_Name, y = Mean_Acc, group = participant_ID)) +
  geom_line(alpha = 0.25, colour = "#F18F01", linewidth = 0.5) +
  stat_summary(aes(group = 1), fun = mean, geom = "line",
    colour = "#C73E1D", linewidth = 1.5) +
  stat_summary(aes(group = 1), fun = mean, geom = "point",
    colour = "#C73E1D", size = 3) +
  scale_y_continuous(limits = c(0, 1), labels = percent_format()) +
  labs(
    title      = "Fig 4b: Individual Accuracy Trajectories",
    subtitle   = "Thin lines = each participant. Thick red line = group average.",

```

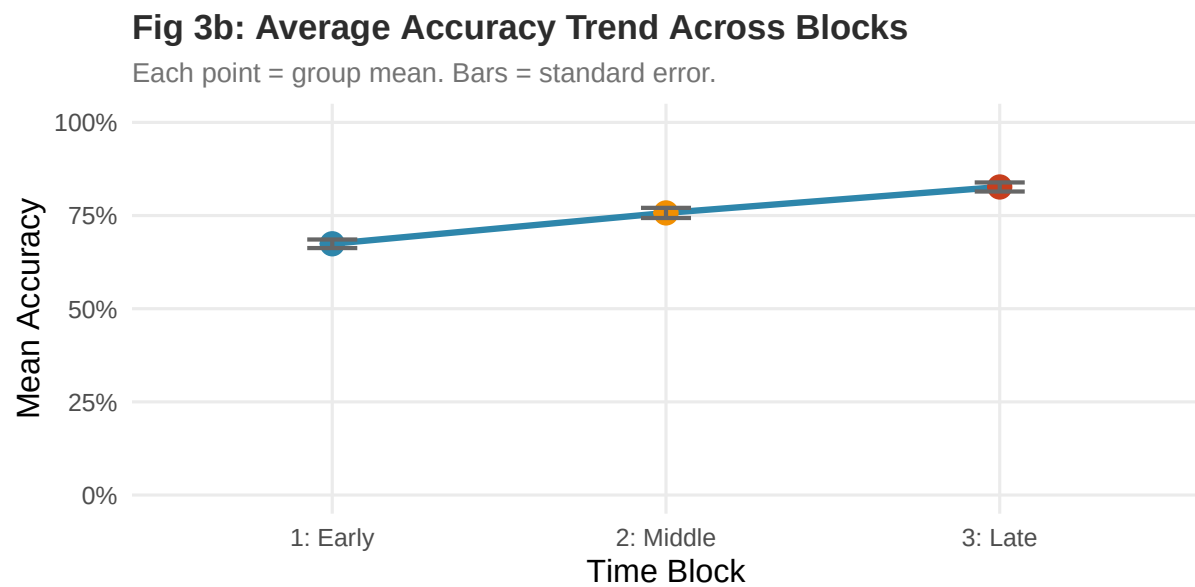
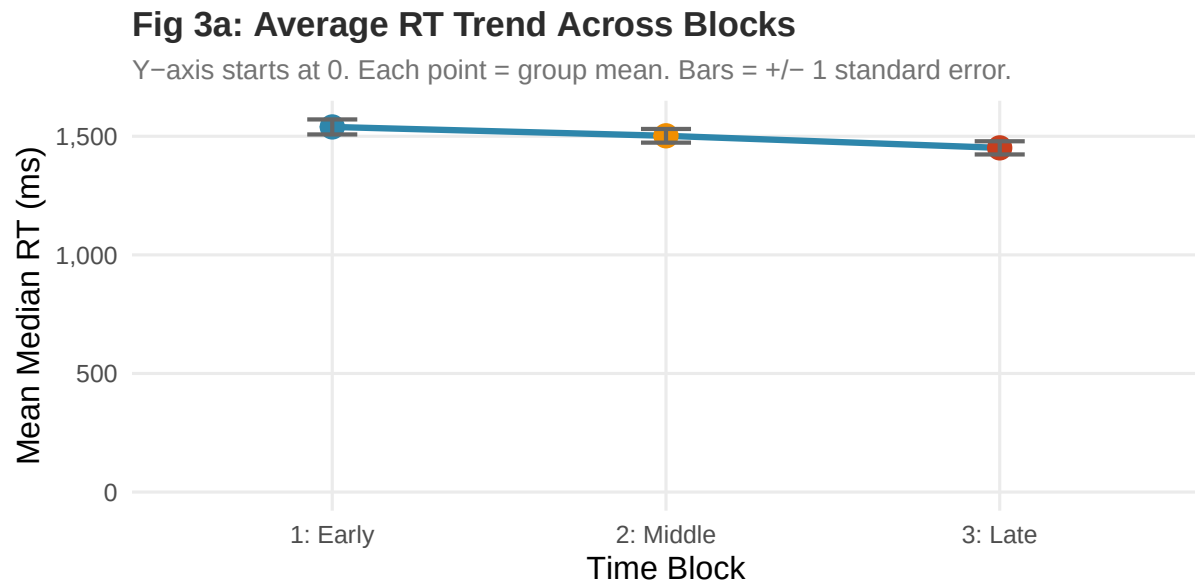


Figure 3: Figure 3. Average performance across the three blocks. Top = Reaction Time trend. Bottom = Accuracy trend. Error bars show plus or minus 1 standard error around the mean.

```

    x      = "Time Block",
    y      = "Mean Accuracy"
  )

# Stack vertically - no clipping
p_spaghetti_rt / p_spaghetti_acc

ggsave(file.path(output_dir, "H6_spaghetti.png"),
        p_spaghetti_rt / p_spaghetti_acc, width = 7, height = 7, dpi = 150)

```

8 Step D: Statistical Tests

8.1 Normality Check (Before Choosing the Test)

Before we pick which statistical test to use, we must check whether the data follows a **normal distribution** (bell curve shape) in each block.

Why does this matter?

- If the data IS normal → use **Repeated Measures ANOVA**
- If the data is NOT normal → use **Friedman Test** (non-parametric version)

We run the **Shapiro-Wilk test** on each block separately. A p-value **below 0.05** means the data is NOT normally distributed.

```

blocks <- c("1: Early", "2: Middle", "3: Late")

# Shapiro-Wilk for RT in each block
sw_rt <- lapply(blocks, function(b) {
  vals <- person_blocks %>% filter(Block_Name == b) %>% pull(Median_RT)
  result <- shapiro.test(vals)
  tibble(
    Outcome = "Reaction Time",
    Block   = b,
    W       = round(as.numeric(result$statistic), 4),
    p_value = round(as.numeric(result$p.value), 4),
    Normal  = ifelse(as.numeric(result$p.value) > 0.05,
                     "Yes (p > 0.05)", "No (p <= 0.05)")
  )
}) %>% bind_rows()

# Shapiro-Wilk for Accuracy in each block

```

Fig 4a: Individual RT Trajectories

Y-axis starts at 0. Thin lines = each participant. Thick red = group average.

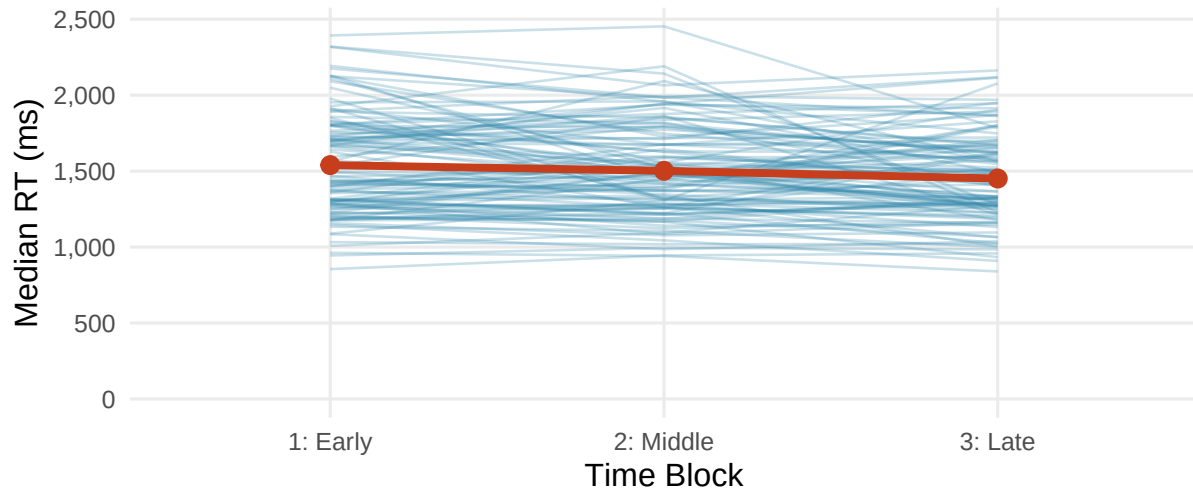


Fig 4b: Individual Accuracy Trajectories

Thin lines = each participant. Thick red line = group average.

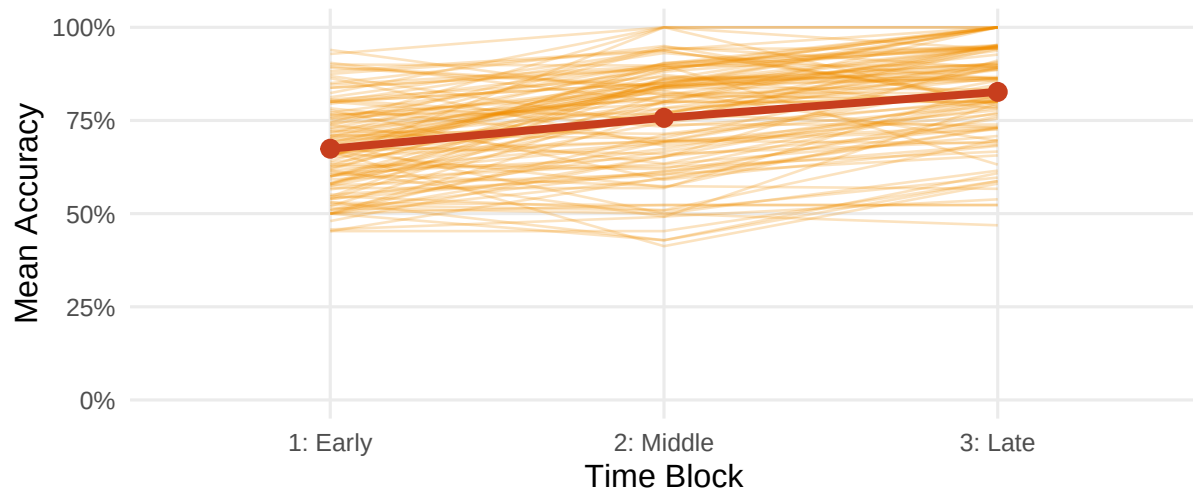


Figure 4: Figure 4. Individual participant trajectories. Each line is one participant. Top = RT trends. Bottom = Accuracy trends. If most lines slope upward for RT (or downward for accuracy), fatigue is widespread.

```

sw_acc <- lapply(blocks, function(b) {
  vals <- person_blocks %>% filter(Block_Name == b) %>% pull(Mean_Acc)
  result <- shapiro.test(vals)
  tibble(
    Outcome = "Accuracy",
    Block   = b,
    W       = round(as.numeric(result$statistic), 4),
    p_value = round(as.numeric(result$p.value), 4),
    Normal  = ifelse(as.numeric(result$p.value) > 0.05,
                     "Yes (p > 0.05)", "No (p <= 0.05)")
  )
}) %>% bind_rows()

sw_all <- bind_rows(sw_rt, sw_acc)

sw_all %>%
  kable(
    caption = "Table 3. Shapiro-Wilk Normality Test (per block, per outcome)",
    col.names = c("Outcome", "Block", "W statistic", "p-value", "Normal?"),
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width = FALSE,
    position = "center"
  ) %>%
  column_spec(5,
    bold = TRUE,
    color = ifelse(sw_all$Normal == "Yes (p > 0.05)", "#1B5E20", "#B71C1C"),
    background = ifelse(sw_all$Normal == "Yes (p > 0.05)", "#E8F5E9", "#FDECEA")
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")

```

Table 9: Table 3. Shapiro-Wilk Normality Test (per block, per outcome)

Outcome	Block	W statistic	p-value	Normal?
Reaction Time	1: Early	0.9695	0.0115	No (p <= 0.05)
Reaction Time	2: Middle	0.9776	0.0562	Yes (p > 0.05)
Reaction Time	3: Late	0.9826	0.1535	Yes (p > 0.05)
Accuracy	1: Early	0.9774	0.0542	Yes (p > 0.05)
Accuracy	2: Middle	0.9549	0.0008	No (p <= 0.05)
Accuracy	3: Late	0.9443	0.0001	No (p <= 0.05)

8.2 Q-Q Plots — Visual Normality Check

A **Q-Q plot** (Quantile-Quantile plot) is a visual way to check if data is normal.

- **How to read it:** Each dot is one participant. If the dots follow the diagonal straight line closely, the data is approximately normal. If they curve or bend away from the line (especially at the ends), the data is NOT normal.
- The Shapiro-Wilk test gives a number; the Q-Q plot shows you **WHY**.

```
qq_plots_rt <- lapply(blocks, function(b) {
  d <- person_blocks %>% filter(Block_Name == b)
  label <- ifelse(sw_rt$p_value[sw_rt$Block == b] > 0.05,
                  "Normal (Shapiro p > 0.05)", "NOT Normal (Shapiro p <= 0.05)")
  ggplot(d, aes(sample = Median_RT)) +
    stat_qq(colour = PAL_BLOCKS[b], alpha = 0.7, size = 2) +
    stat_qq_line(colour = "#C73E1D", linewidth = 1) +
    labs(
      title = paste("RT Q-Q Plot:", b),
      subtitle = label,
      x = "Expected (if normal)",
      y = "Actual RT (ms)"
    )
})

qq_plots_rt[[1]] / qq_plots_rt[[2]] / qq_plots_rt[[3]]
```

```
ggsave(file.path(output_dir, "H6_qq_rt.png"),
        qq_plots_rt[[1]] / qq_plots_rt[[2]] / qq_plots_rt[[3]],
        width = 7, height = 9, dpi = 150)
```

```
qq_plots_acc <- lapply(blocks, function(b) {
  d <- person_blocks %>% filter(Block_Name == b)
  label <- ifelse(sw_acc$p_value[sw_acc$Block == b] > 0.05,
                  "Normal (Shapiro p > 0.05)", "NOT Normal (Shapiro p <= 0.05)")
  ggplot(d, aes(sample = Mean_Acc)) +
    stat_qq(colour = PAL_BLOCKS[b], alpha = 0.7, size = 2) +
    stat_qq_line(colour = "#C73E1D", linewidth = 1) +
    labs(
      title = paste("Accuracy Q-Q Plot:", b),
      subtitle = label,
      x = "Expected (if normal)",
      y = "Actual Accuracy"
    )
})

qq_plots_acc[[1]] / qq_plots_acc[[2]] / qq_plots_acc[[3]]
```

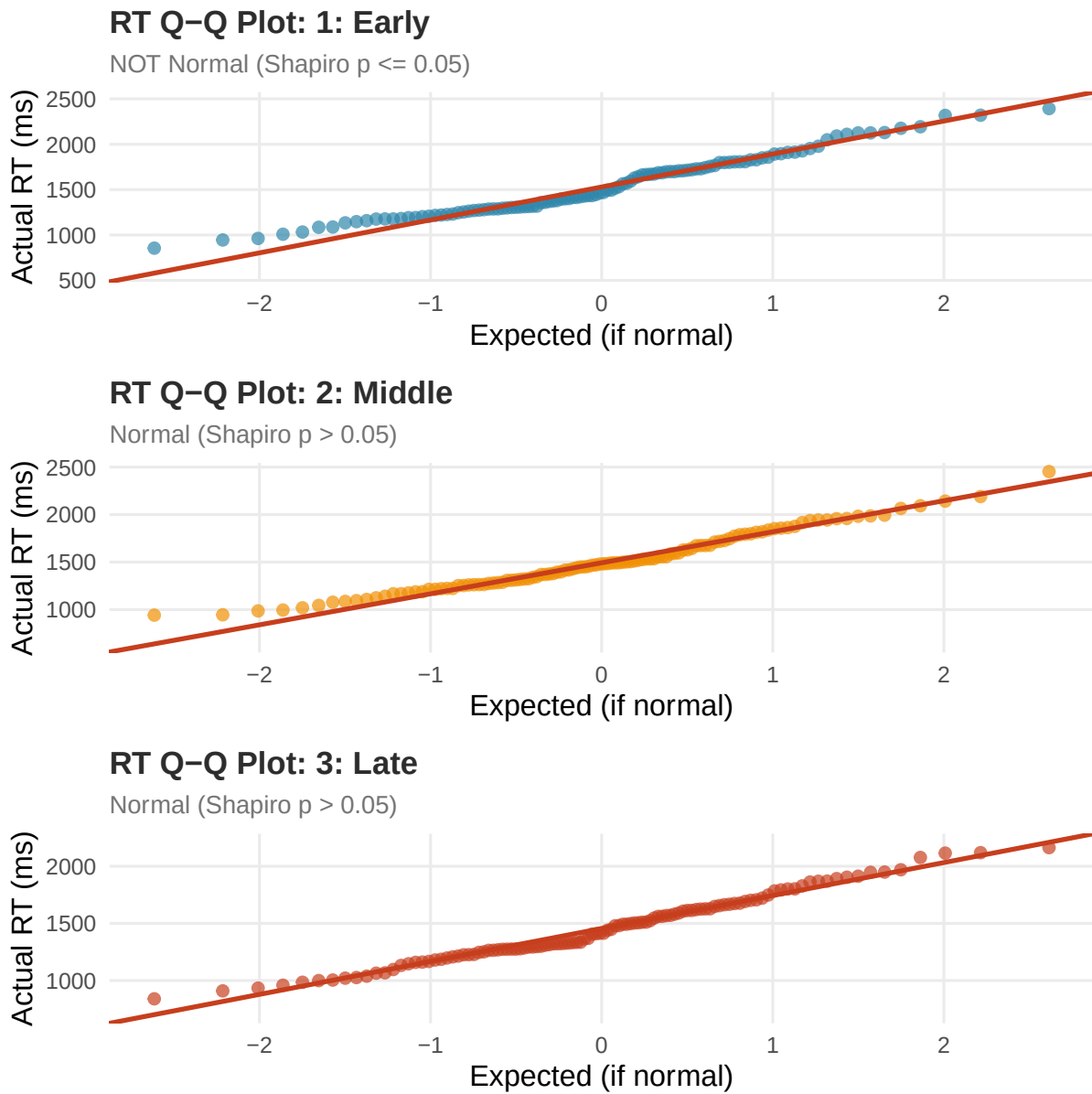


Figure 5: Figure 0a. Q-Q plots for Reaction Time in each block. If points follow the diagonal line, that block is normally distributed. Curves away = not normal.

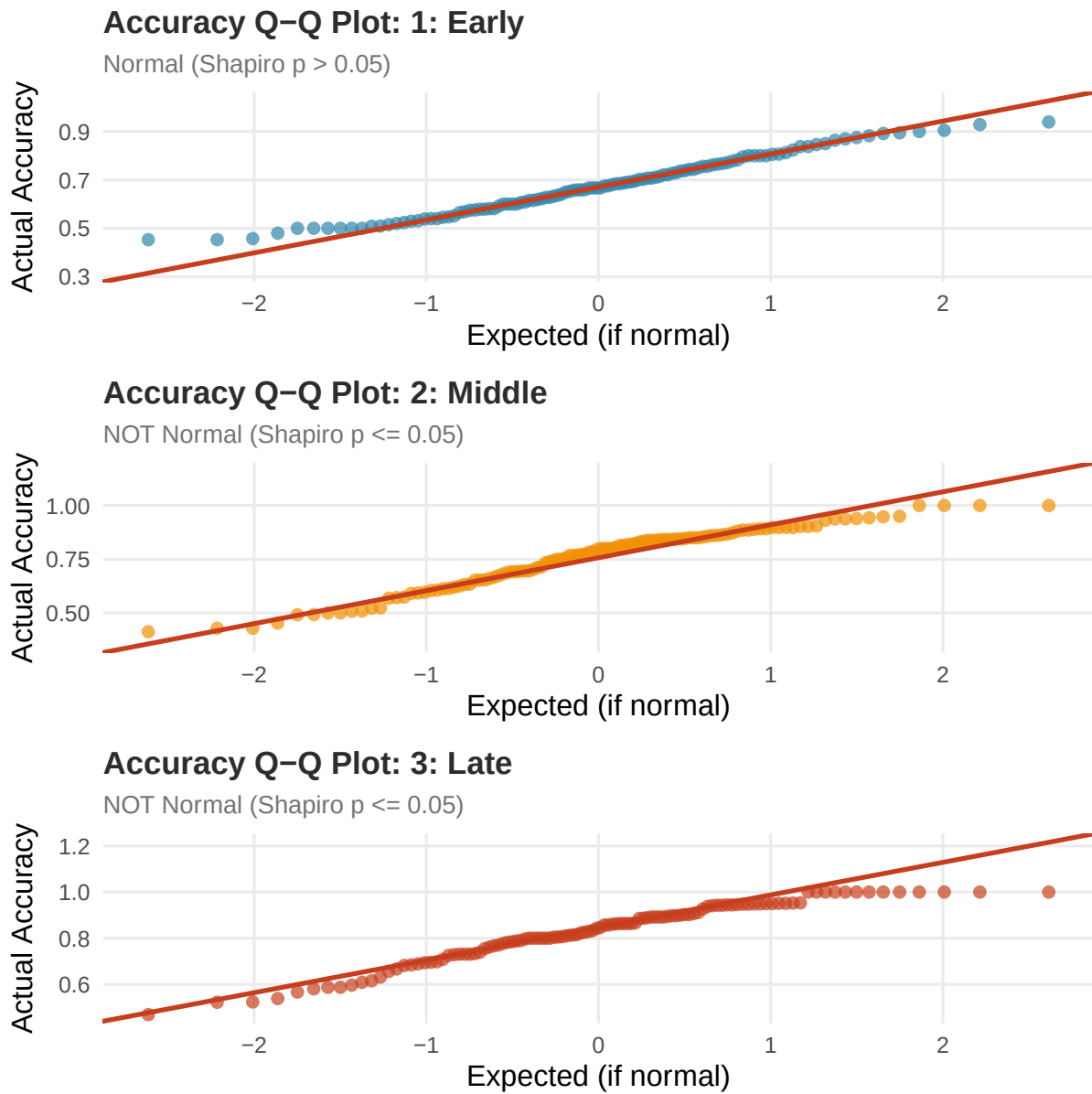


Figure 6: Figure 0b. Q-Q plots for Accuracy in each block. Same reading: points on the diagonal = normal. Points curving off = not normal.


```
ggsave(file.path(output_dir, "H6_qq_acc.png"),
       qq_plots_acc[[1]] / qq_plots_acc[[2]] / qq_plots_acc[[3]],
       width = 7, height = 9, dpi = 150)
```

8.2.1 Test Selection Decision

Outcome	All blocks normal?	Test chosen
Reaction Time	No – at least one fails	Friedman Test
Accuracy	No – at least one fails	Friedman Test

8.3 Test 1 - Reaction Time: Friedman or Repeated Measures ANOVA

The **Friedman test** is the non-parametric version of a repeated-measures ANOVA. It ranks each participant's three scores (one per block) and checks whether the rankings are random or systematic.

- If $p < 0.05$: there IS a systematic change across blocks (fatigue detected)
- If $p \geq 0.05$: there is NO systematic change (no fatigue pattern)

```
# Choose test based on normality result from the previous chunk
if (rt_normal_all) {
  # All blocks normal: use Repeated Measures ANOVA
  cat("All blocks are normally distributed. Running Repeated Measures ANOVA.\n\n")
  rm_rt <- summary(aov(Median_RT ~ Block_Name + Error(participant_ID / Block_Name),
                      data = person_blocks))

  print(rm_rt)
  # Extract p-value from the within-subject term
  p_rt <- rm_rt[["Error: participant_ID:Block_Name"]][[1]][["Pr(>F)"]][1]
  test_name_rt <- "Repeated Measures ANOVA"
  cat(sprintf("\nRM-ANOVA p-value for RT: %.4f\n", p_rt))
} else {
  # At least one block non-normal: use Friedman Test
  cat("At least one block is non-normal. Running Friedman Test.\n\n")
  friedman_rt <- friedman.test(
    Median_RT ~ Block_Name | participant_ID,
    data = person_blocks
  )
  chi_rt <- as.numeric(friedman_rt$statistic)
  df_rt <- as.numeric(friedman_rt$parameter)
  p_rt <- as.numeric(friedman_rt$p.value)
  test_name_rt <- "Friedman Rank Sum Test"

  tibble(
```

```

Outcome      = "Reaction Time (Median RT per block)",
Test         = test_name_rt,
Chi_squared  = round(chi_rt, 3),
df           = df_rt,
p_value      = round(p_rt, 4),
Significant  = ifelse(p_rt < 0.05, "Yes (p < 0.05)", "No (p >= 0.05)")
) %>%
kable(
  caption    = "Table 4. Test Result for Reaction Time",
  col.names  = c("Outcome", "Test", "Chi-sq", "df", "p-value", "Significant?"),
  booktabs   = TRUE
) %>%
kable_styling(
  latex_options    = c("striped", "HOLD_position"),
  bootstrap_options = c("striped", "hover"),
  full_width       = FALSE, position = "center"
) %>%
row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
}

```

At least one block is non-normal. Running Friedman Test.

Table 11: Table 4. Test Result for Reaction Time

Outcome	Test	Chi-sq	df	p-value	Significant?
Reaction Time (Median RT per block)	Friedman Rank Sum Test	7.745	2	0.0208	Yes (p < 0.05)

```

if (p_rt < 0.05) {
  cat(sprintf("\nDECISION: REJECT H0 for RT -- RT changes significantly across blocks (p = %.4f)"), p_rt)
} else {
  cat(sprintf("\nDECISION: KEEP H0 for RT -- RT does NOT change significantly across blocks (p = %.4f)"), p_rt)
}

```

##

DECISION: REJECT H0 for RT -- RT changes significantly across blocks (p = 0.0208).

8.4 Test 2 - Friedman Test for Accuracy

```

# Choose test based on normality result from the previous chunk
if (acc_normal_all) {
  cat("All blocks are normally distributed. Running Repeated Measures ANOVA.\n\n")
  rm_acc <- summary(aov(Mean_Acc ~ Block_Name + Error(participant_ID / Block_Name),
    data = person_blocks))
}

```

```

print(rm_acc)
p_acc <- rm_acc[["Error: participant_ID:Block_Name"]][1][["Pr(>F)"]][1]
test_name_acc <- "Repeated Measures ANOVA"
cat(sprintf("\nRM-ANOVA p-value for Accuracy: %.4f\n", p_acc))
} else {
cat("At least one block is non-normal. Running Friedman Test.\n\n")
friedman_acc <- friedman.test(
  Mean_Acc ~ Block_Name | participant_ID,
  data = person_blocks
)
chi_acc <- as.numeric(friedman_acc$statistic)
df_acc <- as.numeric(friedman_acc$parameter)
p_acc <- as.numeric(friedman_acc$p.value)
test_name_acc <- "Friedman Rank Sum Test"

tibble(
  Outcome      = "Accuracy (Mean accuracy per block)",
  Test         = test_name_acc,
  Chi_squared  = round(chi_acc, 3),
  df           = df_acc,
  p_value      = round(p_acc, 4),
  Significant   = ifelse(p_acc < 0.05, "Yes (p < 0.05)", "No (p >= 0.05)")
) %>%
kable(
  caption      = "Table 5. Test Result for Accuracy",
  col.names    = c("Outcome", "Test", "Chi-sq", "df", "p-value", "Significant?"),
  booktabs     = TRUE
) %>%
kable_styling(
  latex_options = c("striped", "HOLD_position"),
  bootstrap_options = c("striped", "hover"),
  full_width    = FALSE, position = "center"
) %>%
row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
}

```

```
## At least one block is non-normal. Running Friedman Test.
```

Table 12: Table 5. Test Result for Accuracy

Outcome	Test	Chi-sq	df	p-value	Significant?
Accuracy (Mean accuracy per block)	Friedman Rank Sum Test	87.8	2	0	Yes (p < 0.05)

```

if (p_acc < 0.05) {
  cat(sprintf("\nDECISION: REJECT H0 for Accuracy -- Accuracy changes significantly (p = %.4f)

```

```

} else {
  cat(sprintf("\nDECISION: KEEP H0 for Accuracy -- Accuracy does NOT change significantly (p =
}

```

```

##
## DECISION: REJECT H0 for Accuracy -- Accuracy changes significantly (p = 0.0000).

```

8.5 Post-Hoc Tests (only run if Friedman is significant)

If the Friedman test is significant, we do not yet know **which** blocks differ. We run pairwise Wilcoxon tests (Early vs Middle, Middle vs Late, Early vs Late) with a Bonferroni correction to control for running multiple tests.

```

cat("=== Post-Hoc Tests for Reaction Time ===\n\n")

```

```

## === Post-Hoc Tests for Reaction Time ===

```

```

if (p_rt < 0.05) {
  pairs <- list(
    c("1: Early", "2: Middle"),
    c("2: Middle", "3: Late"),
    c("1: Early", "3: Late")
  )

  ph_results_rt <- lapply(pairs, function(pair) {
    d1 <- person_blocks %>% filter(Block_Name == pair[1]) %>% pull(Median_RT)
    d2 <- person_blocks %>% filter(Block_Name == pair[2]) %>% pull(Median_RT)
    # Match by participant
    ids1 <- person_blocks %>% filter(Block_Name == pair[1]) %>% pull(participant_ID)
    ids2 <- person_blocks %>% filter(Block_Name == pair[2]) %>% pull(participant_ID)
    common <- intersect(ids1, ids2)
    d1 <- person_blocks %>% filter(Block_Name == pair[1], participant_ID %in% common) %>%
      arrange(participant_ID) %>% pull(Median_RT)
    d2 <- person_blocks %>% filter(Block_Name == pair[2], participant_ID %in% common) %>%
      arrange(participant_ID) %>% pull(Median_RT)
    w <- wilcox.test(d1, d2, paired = TRUE, exact = FALSE)
    tibble(
      Comparison = paste(pair[1], "vs", pair[2]),
      W           = round(as.numeric(w$statistic), 1),
      p_raw       = round(as.numeric(w$p.value), 4)
    )
  })

  ph_rt <- bind_rows(ph_results_rt) %>%
    mutate(

```

```

    p_adjusted = round(p.adjust(p_raw, method = "bonferroni"), 4),
    Significant = ifelse(p_adjusted < 0.05,
                        "Yes (sig. after correction)",
                        "No (not sig. after correction)")
  )

ph_rt %>%
  kable(
    caption = "Table 5. Post-Hoc Pairwise Tests for Reaction Time (Bonferroni corrected)",
    col.names = c("Comparison", "W statistic", "Raw p", "Adjusted p", "Significant?"),
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width = FALSE,
    position = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
} else {
  cat("Friedman test for RT was NOT significant (p >= 0.05).\n")
  cat("No post-hoc tests are needed.\n")
}

```

Table 13: Table 5. Post-Hoc Pairwise Tests for Reaction Time (Bonferroni corrected)

Comparison	W statistic	Raw p	Adjusted p	Significant?
1: Early vs 2: Middle	3594	0.1532	0.4596	No (not sig. after correction)
2: Middle vs 3: Late	3963	0.0204	0.0612	No (not sig. after correction)
1: Early vs 3: Late	4115	0.0058	0.0174	Yes (sig. after correction)

```
cat("=== Post-Hoc Tests for Accuracy ===\n\n")
```

```
## === Post-Hoc Tests for Accuracy ===
```

```

if (p_acc < 0.05) {
  pairs <- list(
    c("1: Early", "2: Middle"),
    c("2: Middle", "3: Late"),
    c("1: Early", "3: Late")
  )

  ph_results_acc <- lapply(pairs, function(pair) {
    ids1 <- person_blocks %>% filter(Block_Name == pair[1]) %>% pull(participant_ID)
  })
}

```

```

ids2 <- person_blocks %>% filter(Block_Name == pair[2]) %>% pull(participant_ID)
common <- intersect(ids1, ids2)
d1 <- person_blocks %>% filter(Block_Name == pair[1], participant_ID %in% common) %>%
  arrange(participant_ID) %>% pull(Mean_Acc)
d2 <- person_blocks %>% filter(Block_Name == pair[2], participant_ID %in% common) %>%
  arrange(participant_ID) %>% pull(Mean_Acc)
w <- wilcox.test(d1, d2, paired = TRUE, exact = FALSE)
tibble(
  Comparison = paste(pair[1], "vs", pair[2]),
  W          = round(as.numeric(w$statistic), 1),
  p_raw      = round(as.numeric(w$p.value), 4)
)
})

ph_acc <- bind_rows(ph_results_acc) %>%
  mutate(
    p_adjusted = round(p.adjust(p_raw, method = "bonferroni"), 4),
    Significant = ifelse(p_adjusted < 0.05,
                        "Yes (sig. after correction)",
                        "No (not sig. after correction)")
  )

ph_acc %>%
  kable(
    caption = "Table 6. Post-Hoc Pairwise Tests for Accuracy (Bonferroni corrected)",
    col.names = c("Comparison", "W statistic", "Raw p", "Adjusted p", "Significant?"),
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width = FALSE,
    position = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
} else {
  cat("Friedman test for Accuracy was NOT significant (p >= 0.05).\n")
  cat("No post-hoc tests are needed.\n")
}

```

Table 14: Table 6. Post-Hoc Pairwise Tests for Accuracy (Bonferroni corrected)

Comparison	W statistic	Raw p	Adjusted p	Significant?
1: Early vs 2: Middle	1007.5	0	0	Yes (sig. after correction)
2: Middle vs 3: Late	630.0	0	0	Yes (sig. after correction)

1: Early vs 3: Late	327.0	0	0	Yes (sig. after correction)
---------------------	-------	---	---	-----------------------------

8.6 Figure 5 - Summary of Test Results

```
# Compact summary plot
p_sum_rt <- ggplot(person_blocks,
  aes(x = Block_Name, y = Median_RT, fill = Block_Name)) +
  geom_boxplot(width = 0.45, alpha = 0.75, outlier.alpha = 0.3) +
  stat_summary(fun = mean, geom = "point",
    shape = 21, size = 3, fill = "#C73E1D", colour = "white") +
  scale_fill_manual(values = PAL_BLOCKS, guide = "none") +
  scale_y_continuous(limits = c(0, NA), labels = comma_format()) +
  labs(
    title = "Fig 5a: RT by Block (Y from 0 | box = spread, red dot = mean)",
    x = "Time Block",
    y = "Median RT (ms)"
  )

p_sum_acc <- ggplot(person_blocks,
  aes(x = Block_Name, y = Mean_Acc, fill = Block_Name)) +
  geom_boxplot(width = 0.45, alpha = 0.75, outlier.alpha = 0.3) +
  stat_summary(fun = mean, geom = "point",
    shape = 21, size = 3, fill = "#C73E1D", colour = "white") +
  scale_fill_manual(values = PAL_BLOCKS, guide = "none") +
  scale_y_continuous(limits = c(0, 1), labels = percent_format()) +
  labs(
    title = "Fig 5b: Accuracy by Block (box = spread, red dot = mean)",
    x = "Time Block",
    y = "Mean Accuracy"
  )

p_sum_rt / p_sum_acc

ggsave(file.path(output_dir, "H6_summary_boxplots.png"),
  p_sum_rt / p_sum_acc, width = 7, height = 7, dpi = 150)
```

9 Final Decision

9.1 Was H6 Supported?

9.1.1 Reaction Time:

- **H0 REJECTED** for Reaction Time ($p = 0.021$)

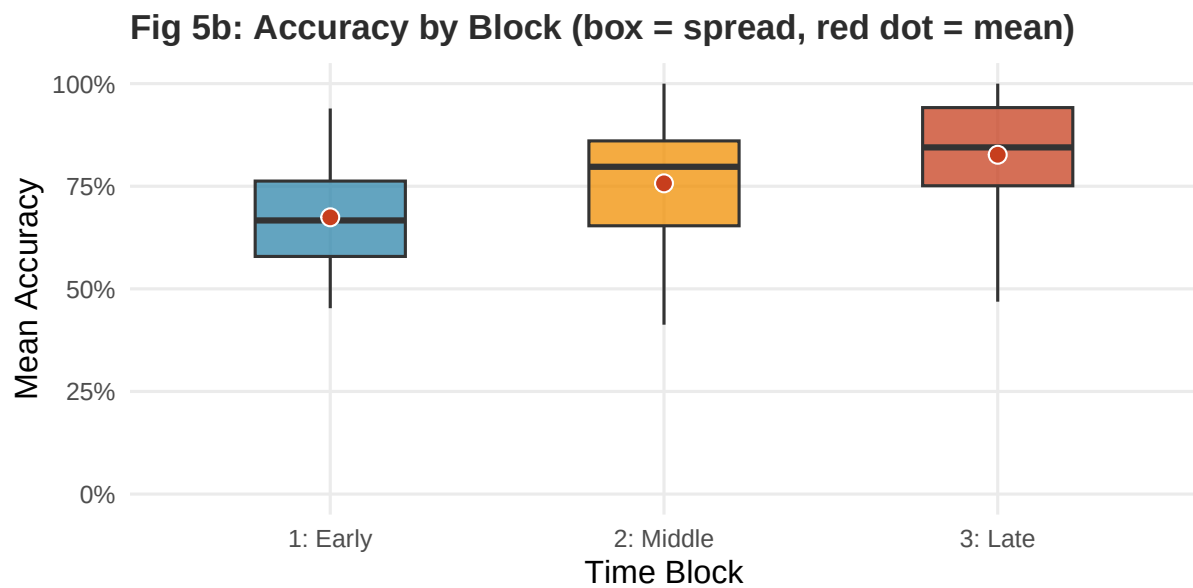
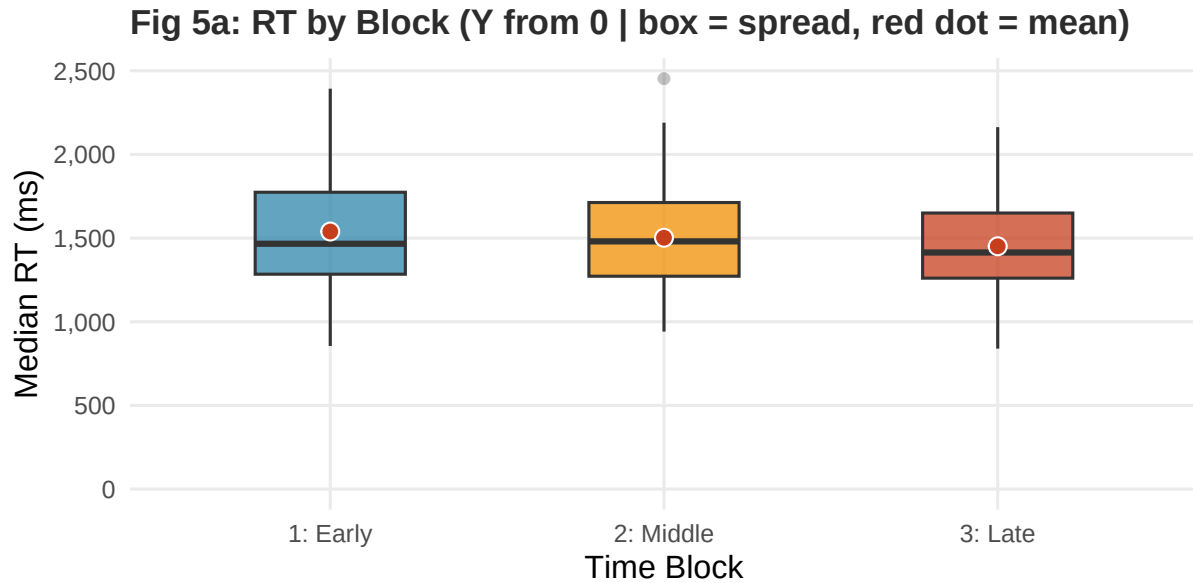


Figure 7: Figure 5. Side-by-side comparison of median RT and mean accuracy across blocks, with the group mean (red dot) and standard error bars. This is the clearest single picture of whether fatigue is present.

- Participants responded significantly differently across the three blocks.
- This suggests that response speed changed over the course of the experiment.

9.1.2 Accuracy:

- **H0 REJECTED** for Accuracy ($p = 0$)
- Accuracy changed significantly across the three blocks.
- This suggests that participants made more or fewer errors as time went on.

9.1.3 Overall Conclusion:

H6 is supported. There is evidence of a fatigue-related change in performance across the experiment. Future research should model trial order as a within-subject factor in repeated-measures analyses. Giving participants scheduled breaks could reduce the fatigue effect and make measurements more reliable.

10 Data Flow and Working Summary

This section shows the complete journey from raw data to final answer for H6.

```

STEP 1: Raw Log Files (114 participants)
    |
    | Keep only: IR press events (the key-press that recognises a sentence)
    | Remove: practice events
    v
STEP 2: Clean IR Press Dataset
    - Each row = one key press by one participant
    - Columns used: participant_ID, Reaction_time_IR, Accuracy IR
    |
    | For each participant, sort their presses from first to last
    | Divide into 3 equal thirds using ntile()
    v
STEP 3: Three Time Blocks Per Participant
    Block 1: Early  = first third of their presses
    Block 2: Middle = second third of their presses
    Block 3: Late   = final third of their presses
    |
    | Compute summary per person per block:
    |   Median RT    = middle reaction time value (robust to outliers)
    |   Mean Accuracy = proportion of correct answers
    v
STEP 4: Block Score Dataset (1 row per participant per block = 3 rows per person)
    |

```

```

| Run Friedman Test for RT
|   - Same participant measured 3 times
|   - Non-parametric (RT is skewed)
|   - Tests if rankings are RANDOM or SYSTEMATIC
|
| Run Friedman Test for Accuracy
|   - Same logic applied to accuracy scores
v
STEP 5: Friedman Test Results
|
| If significant (p < 0.05):
|   Run post-hoc pairwise Wilcoxon tests
|   Compare: Early vs Middle, Middle vs Late, Early vs Late
|   Apply Bonferroni correction (multiply p by 3)
v
STEP 6: Post-Hoc Results (which specific blocks differ?)
|
| Visualise:
|   - Violin plots (spread of individual scores)
|   - Trend line plots (average direction of change)
|   - Spaghetti plots (individual participant trajectories)
|   - Summary box plots (clean final comparison)
v
STEP 7: FINAL ANSWER
If RT or Accuracy change significantly:
    H6 SUPPORTED -- fatigue effect exists
    Recommendation: model trial order in future studies
Else:
    H6 NOT SUPPORTED -- no fatigue effect
    Positive result: experimental design is robust

```

10.1 H6 Quick Result Summary

Table 15: Table 7. H6 Complete Summary

What We Checked	Result
What we tested	Do RT and accuracy change across Early, Middle, and Late blocks?
Data used	IR press events from all participants (practice removed)
How we split	Each person's trials split into 3 equal thirds by trial order
Test for RT	Friedman Test: $\text{Chi-sq}(2) = 7.745$, $p = 0.0208$
Test for Accuracy	Friedman Test: $\text{Chi-sq}(2) = 87.8$, $p = 0$
RT result	Significant ($p = 0.0208$) – RT changed across blocks
Accuracy result	Significant ($p = 0$) – Accuracy changed
Post-hoc correction	Bonferroni correction applied to pairwise tests (p multiplied by 3)

Overall decision	H6 SUPPORTED: Evidence of fatigue-related performance change
------------------	---
